Coordinates on the plane; the midpoint of a segment

Two numbers fix a point — and from there geometry can be done with arithmetic.

LESSON 33–34 | 2 × 40 MIN | GEOMETRY 8, TEMA 31 | STAGE 9 · 3.1

LESSON CLOCK

12' | 20' | 24' | 16' | 8'

Plot five points from dictation	12 min
Reading the plane	20 min
The midpoint formula	24 min
Problems both ways	16 min
Homework	8 min

LEARNING OBJECTIVES

- Plot and read points in all four quadrants.
- Find the midpoint of a segment from the coordinates of its ends.
- Find an endpoint when the midpoint and the other end are known.
- Use coordinates to show that a quadrilateral is a parallelogram.

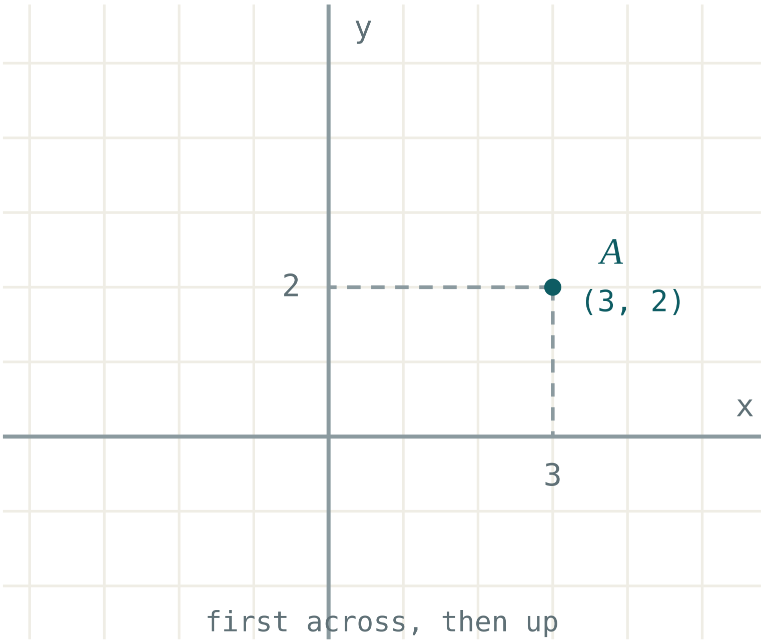
TEXTBOOK

National	Тема 31, pp. 69–71
Cambridge	Learner's Book pp. 56–60
Workbook	Workbook 3.1

01 Explanation

A point is a pair of numbers

Two perpendicular axes meet at the **origin** $O(0; 0)$. Any point of the plane is then fixed by an **ordered pair** $A(x; y)$: first go x along, then y up.



A point is fixed by two numbers — how far across, then how far up. The order matters.

The order matters absolutely: $(3; 5)$ and $(5; 3)$ are different points. The axes cut the plane into four **quadrants**:

QUADRANT	*I*	*II*	*III*	*IV*
x	+	−	−	+
y	+	+	−	−

A point on the x-axis has $y = 0$; a point on the y-axis has $x = 0$. The origin is the only point on both.

The midpoint

MIDPOINT OF A SEGMENT

If $A(x_1; y_1)$ and $B(x_2; y_2)$, the midpoint M of AB is

$$M\left(\frac{x_1 + x_2}{2}\ ;\ \frac{y_1 + y_2}{2}\right)$$

— the average of the two x's and the average of the two y's.

Why the average? Going from A to B the x-coordinate changes by $x_2 - x_1$; halfway there it has changed by half of that:

$$x_1 + \frac{x_2 - x_1}{2} = \frac{2x_1 + x_2 - x_1}{2} = \frac{x_1 + x_2}{2}$$

The same argument works for y. So the formula is not something to memorise blindly — it says “stand halfway”.

READ THE QUESTION

“Find the midpoint” means average. “Find the other end, given the midpoint” means the reverse: $x_2 = 2x\_M - x_1$. Doubling and subtracting, not averaging.

Why coordinates help

Once points are numbers, statements about shapes become statements about arithmetic. Take $A(1; 2)$, $B(5; 3)$, $C(6; 6)$, $D(2; 5)$. Is $ABCD$ a parallelogram?

$$\text{midpoint of } AC\text{: } (3.5; 4) \;\mid\; \text{midpoint of } BD\text{: } (3.5; 4)$$

The diagonals share a midpoint, so they bisect each other — and by the test proved in Quarter I, $ABCD$ is a parallelogram. No drawing, no measuring, no doubt.

This is the **coordinate method**: translate the geometry into coordinates, do the algebra, translate the answer back. The rest of this chapter is built on it.

02 Worked examples

EXAMPLE 1 **Find the midpoint of $A(-3; 4)$ and $B(7; -2)$.**

1. $$x_M = \frac{-3 + 7}{2} = 2$$

   Average the x-coordinates.

2. $$y_M = \frac{4 + (-2)}{2} = 1$$

   Average the y-coordinates.

3. $M(2; 1)$

   Check: it lies between the two, as it must.

ANSWER

$M(2; 1)$

EXAMPLE 2 **$M(4; -1)$ is the midpoint of AB and $A(1; 3)$. Find B.**

1. $$\frac{1 + x}{2} = 4$$

   Write the midpoint formula and solve for x.

2. $1 + x = 8, x = 7$

   Double, then subtract.

3. $$\frac{3 + y}{2} = -1$$

4. $3 + y = -2, y = -5$

5. $B(7; -5)$

   Check the midpoint of $(1;3)$ and $(7;-5)$ is $(4;-1)$ ✓

ANSWER

$B(7; -5)$

EXAMPLE 3 **Show that $A(0; 0)$, $B(4; 1)$, $C(5; 5)$, $D(1; 4)$ form a parallelogram.**

1. Find the midpoint of each diagonal.
   The bisection test.
2. $AC: (2.5; 2.5)$
3. $BD: (2.5; 2.5)$
   The same point.
4. The diagonals bisect each other, so $ABCD$ is a parallelogram.
   Test proved in Quarter I.

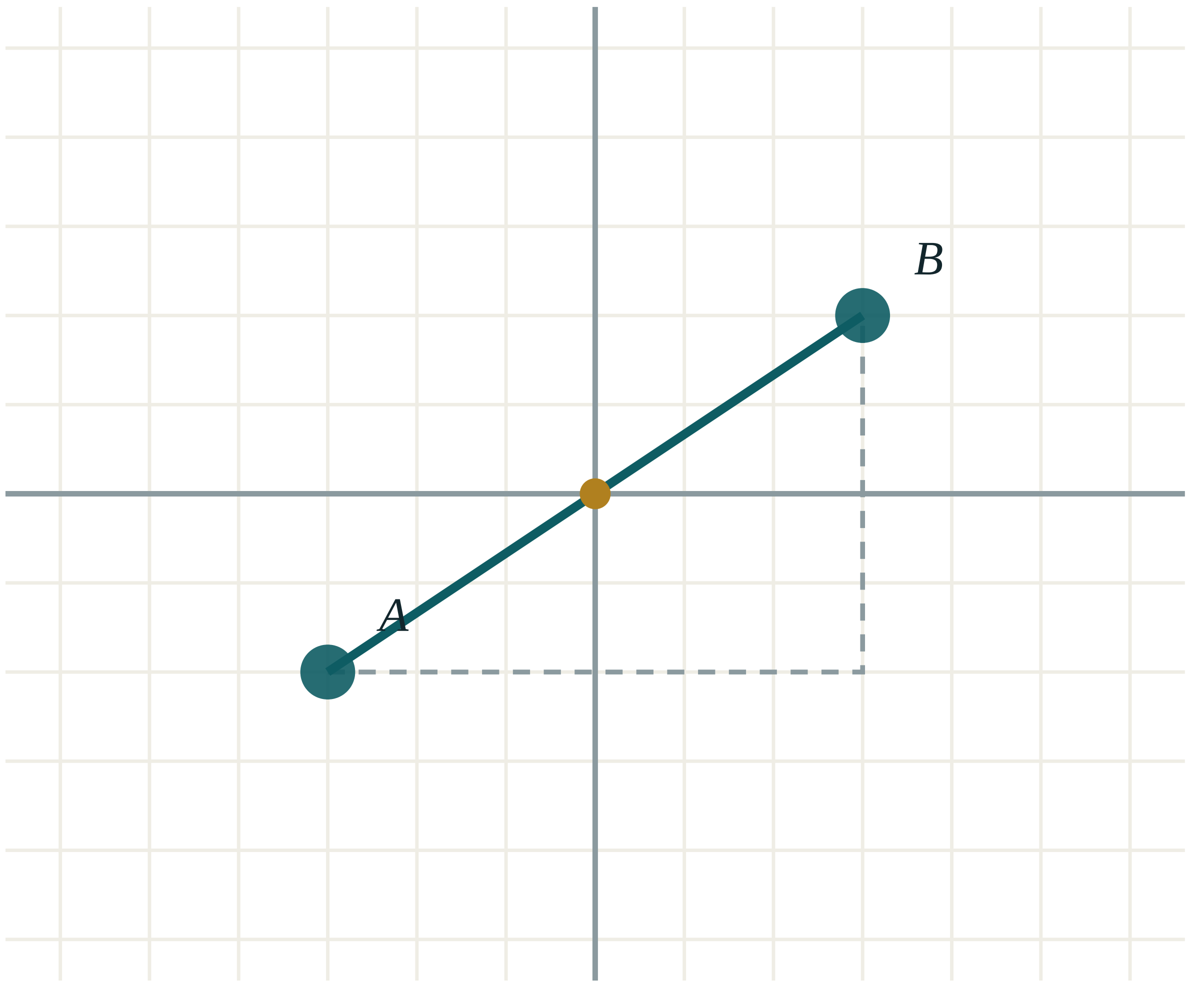
ANSWER

Yes — the diagonals share the midpoint $(2.5; 2.5)$

03 Interactive model

Drag A and B; the golden dot is always the midpoint, and its coordinates are the two averages.

Two points and their midpoint



A **(-3, -2)** | B **(3, 2)** | $x_2 - x_1$ **6** | $y_2 - y_1$ **4** | distance AB **7.21** | midpoint M **(0, 0)**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Coordinate plane	Koordinatalar tekisligi	Координатная плоскость
Origin	Koordinata boshi	Начало координат
Abscissa (x-coordinate)	Abssissa	Абсцисса
Ordinate (y-coordinate)	Ordinata	Ордината
Quadrant	Chorak	Координатная четверть
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Midpoint	O‘rta nuqta	Середина отрезка
Segment	Kesma	Отрезок
Axis	O‘q	Ось
Coordinate method	Koordinatalar usuli	Метод координат

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The midpoint of $(2; 6)$ and $(8; 2)$ is:

- $(5; 4)$
- $(10; 8)$
- $(6; 4)$
- $(3; 2)$

$(-4; 3)$ lies in quadrant:

- I
- II
- III
- IV

If $M(3; 5)$ is the midpoint of $A(1; 2)B$, then B is:

- $(2; 3.5)$
- $(5; 8)$
- $(4; 7)$
- $(6; 10)$

A point on the y-axis has:

- $y = 0$
- $x = 0$
- $x = y$
- $x = 1$

The midpoint of a segment from $(-5; -3)$ to $(5; 3)$ is:

- $(0; 0)$
- $(5; 3)$
- $(0; 3)$
- $(10; 6)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the midpoint of (0; 0) and (6; 8).*

   ANSWER *(3; 4)*

2. *Find the midpoint of (2; 5) and (8; 11).*

   ANSWER *(5; 8)*

3. *In which quadrant is (3; −7)?*

   ANSWER *IV*

4. *Find the midpoint of (−2; 4) and (6; 4).*

   ANSWER *(2; 4)*

5. *Write the coordinates of the origin.*

   ANSWER *(0; 0)*

6. *Find the midpoint of (−1; −1) and (5; 7).*

   ANSWER *(2; 3)*

7. *A point has $x = 0$ and $y = -4$. Where does it lie?*

   ANSWER On the y-axis, below the origin.

Medium · the standard question

7 PROBLEMS

1. *Find the midpoint of* $(-7; 2)$ *and* $(3; -8)$.

   ANSWER $(-2; -3)$

2. $M(2; 3)$ *is the midpoint of* $A(-1; 5)B$. *Find* B.

   ANSWER $B(5; 1)$

3. $M(0; 0)$ *is the midpoint of* $A(-4; 6)B$. *Find* B.

   ANSWER $B(4; -6)$

4. *Find the midpoint of* $(1.5; 2.5)$ *and* $(4.5; 7.5)$.

   ANSWER $(3; 5)$

5. *The midpoints of the diagonals of* $ABCD$ *are* $(2; 3)$ *and* $(2; 3)$. *What can you conclude?*

   ANSWER The diagonals bisect each other, so $ABCD$ is a parallelogram.

6. $A(1; 1)$, $B(7; 3)$. *Find the point one quarter of the way from* A *to* B.

   ANSWER $(2.5; 1.5)$ — the midpoint of A and the midpoint of AB

7. *Find the midpoint of the segment joining* $(a; b)$ *and* $(3a; 5b)$.

   ANSWER $(2a; 3b)$

Hard · stretch and reason

7 PROBLEMS

1. *$A(1; 2)$, $B(5; 4)$, $C(7; 8)$. Find D so that $ABCD$ is a parallelogram.*

   ANSWER $D(3; 6)$ — the diagonals must share the midpoint $(4; 5)$

2. *Show that $(0; 0)$, $(6; 2)$, $(8; 8)$, $(2; 6)$ form a parallelogram.*

   ANSWER Both diagonals have midpoint $(4; 4)$.

3. *The vertices of a triangle are $(0; 0)$, $(6; 0)$, $(2; 8)$. Find the midpoints of all three sides.*

   ANSWER $(3; 0)$, $(4; 4)$, $(1; 4)$

4. *M is the midpoint of AB, with $A(-3; 7)$ and $M(1; 1)$. Find B and the midpoint of AM.*

   ANSWER $B(5; -5)$, midpoint of AM is $(-1; 4)$

5. *$P(2; 3)$ and $Q(10; 7)$. Find the two points that divide PQ into three equal parts.*

   ANSWER $(4\frac{2}{3}; 4\frac{1}{3})$ and $(7\frac{1}{3}; 5\frac{2}{3})$

6. *Prove that the diagonals of the quadrilateral $(0;0)$, $(a;0)$, $(a+b;c)$, $(b;c)$ bisect each other.*

   ANSWER Both diagonals have midpoint $(\frac{a+b}{2}; \frac{c}{2})$, so the figure is a parallelogram for all a, b, c.

7. *$A(1; 1)$ and $B(9; 5)$. A point C on AB has $AC = 3 \cdot CB$. Find C.*

   ANSWER $(7; 4)$

07 Homework

Homework — 6 problems

Geometry 8, Тема 31, pp. 69–71. Sketch the axes for every problem, even the easy ones.

1. *Find the midpoint of $(3; 7)$ and $(11; 1)$.*
2. *Find the midpoint of $(-6; 2)$ and $(4; -8)$.*
3. *$M(3; -2)$ is the midpoint of $A(7; 4)B$. Find B.*
4. *Plot $A(-2; 3)$, $B(4; 3)$, $C(4; -1)$, $D(-2; -1)$ and name the shape.*

5. *Show that (1; 1), (5; 2), (6; 6), (2; 5) form a parallelogram.*

6. *Find the midpoints of all three sides of the triangle (0; 0), (8; 0), (4; 6).*

The distance between two points; the equation of a circle

Pythagoras, written in coordinates — and the equation that a circle turns out to be.

LESSON 35–36 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 32–33 | STAGE 9 · 3.1, 7.1

LESSON CLOCK

12' | 24' | 24' | 12' | 8'

Recall Pythagoras	12 min
The distance formula	24 min
The equation of a circle	24 min
Inside, on or outside	12 min
Homework	8 min

LEARNING OBJECTIVES

- Find the distance between two points from their coordinates.
- Recognise $(x - a)^2 + (y - b)^2 = r^2$ as the equation of a circle.
- Write the equation of a circle from its centre and radius, and read them back.
- Decide whether a given point lies inside, on or outside a circle.

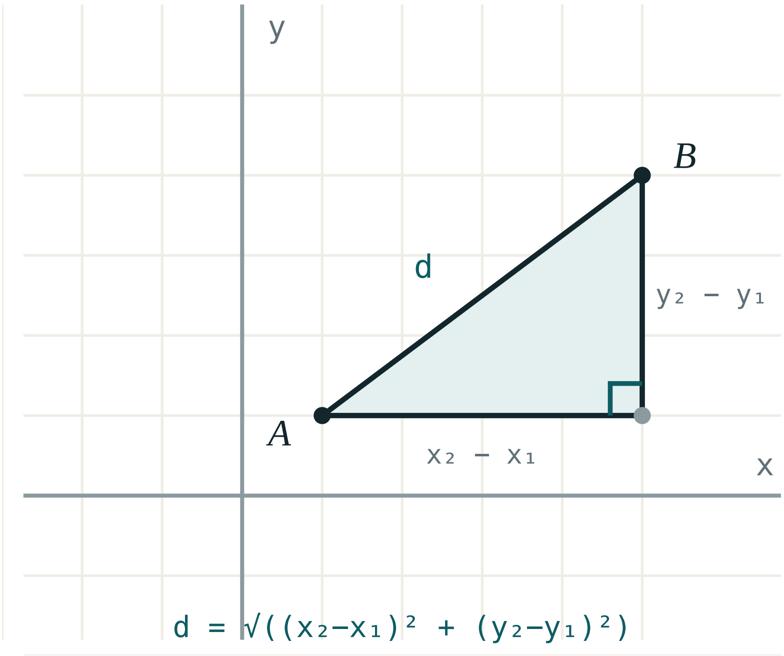
TEXTBOOK

National	Темы 32–33, pp. 72–74
Cambridge	Learner's Book pp. 56–60, 142–148
Workbook	Workbook 3.1

01 Explanation

Distance is Pythagoras in disguise

Join $A(x_1; y_1)$ to $B(x_2; y_2)$ and complete a right-angled triangle with legs parallel to the axes. The horizontal leg is $x_2 - x_1$, the vertical leg is $y_2 - y_1$, and the segment AB is the hypotenuse.



The distance between two points is the hypotenuse of a right triangle whose legs are the differences of the coordinates.

DISTANCE FORMULA

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The squares make the order of subtraction irrelevant: $(3 - 7)^2 = (7 - 3)^2 = 16$. That is the one place where this formula forgives you — everywhere else, be careful.

Leave the answer in surd form unless a decimal is asked for. $\sqrt{20} = 2\sqrt{5}$ is exact; 4.47 is not.

What a circle is, as an equation

A circle of radius r centred at $C(a; b)$ is the set of all points at distance exactly r from C. Write that with the distance formula:

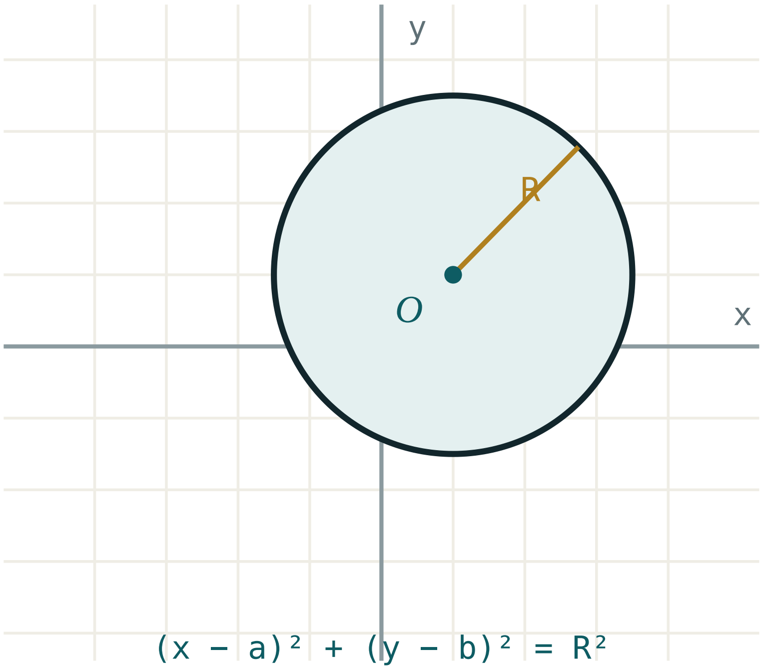
$$\sqrt{(x - a)^2 + (y - b)^2} = r$$

Square both sides — both are positive, so nothing is lost:

EQUATION OF A CIRCLE

$$(x - a)^2 + (y - b)^2 = r^2$$

Centred at the origin this becomes simply $x^2 + y^2 = r^2$.



Every point of the circle is at the same distance r from the centre — that single condition is the equation.

READ THE SIGNS BACKWARDS

$(x - 3)^2 + (y + 2)^2 = 25$ has centre $(3; -2)$, not $(3; 2)$, and radius 5, not 25. The formula subtracts the centre, so the sign you see is the opposite of the sign you want.

Inside, on, or outside

Compare the distance from the centre with the radius:

CONDITION	POSITION OF THE POINT
$(x - a)^2 + (y - b)^2 < r^2$	inside the circle
$(x - a)^2 + (y - b)^2 = r^2$	on the circle
$(x - a)^2 + (y - b)^2 > r^2$	outside the circle

There is no need to take a square root — just substitute the point into the left-hand side and compare with r^2. Is $(6; 1)$ inside $x^2 + y^2 = 36$? Substituting gives $36 + 1 = 37 > 36$, so it is just outside.

02 Worked examples

EXAMPLE 1 **Find the distance between $A(1; 2)$ and $B(5; 5)$.**

1. $x_2 - x_1 = 4$

   The horizontal leg.

2. $y_2 - y_1 = 3$

   The vertical leg.

3. $AB = \sqrt{16 + 9} = \sqrt{25}$

4. $AB = 5$

   A 3–4–5 triangle.

ANSWER

5

EXAMPLE 2 **Write the equation of the circle with centre $(-2; 3)$ and radius 4.**

1. $a = -2, b = 3, r = 4$

2. $(x - (-2))^2 + (y - 3)^2 = 4^2$

   Substitute into the standard form.

3. $(x + 2)^2 + (y - 3)^2 = 16$

   Two minus signs make a plus.

ANSWER

$(x + 2)^2 + (y - 3)^2 = 16$

EXAMPLE 3

A circle has equation $(x - 1)^2 + (y + 4)^2 = 25$. State its centre and radius, and say whether $(4; 0)$ lies on it.

1. *centre* $(1; -4)$

   Change the sign of what is subtracted.

2. $r = 5$

   $25 = 5^2$.

3. $(4 - 1)^2 + (0 + 4)^2 = 9 + 16 = 25$

   Substitute the point.

4. $25 = r^2$

   So the point lies exactly on the circle.

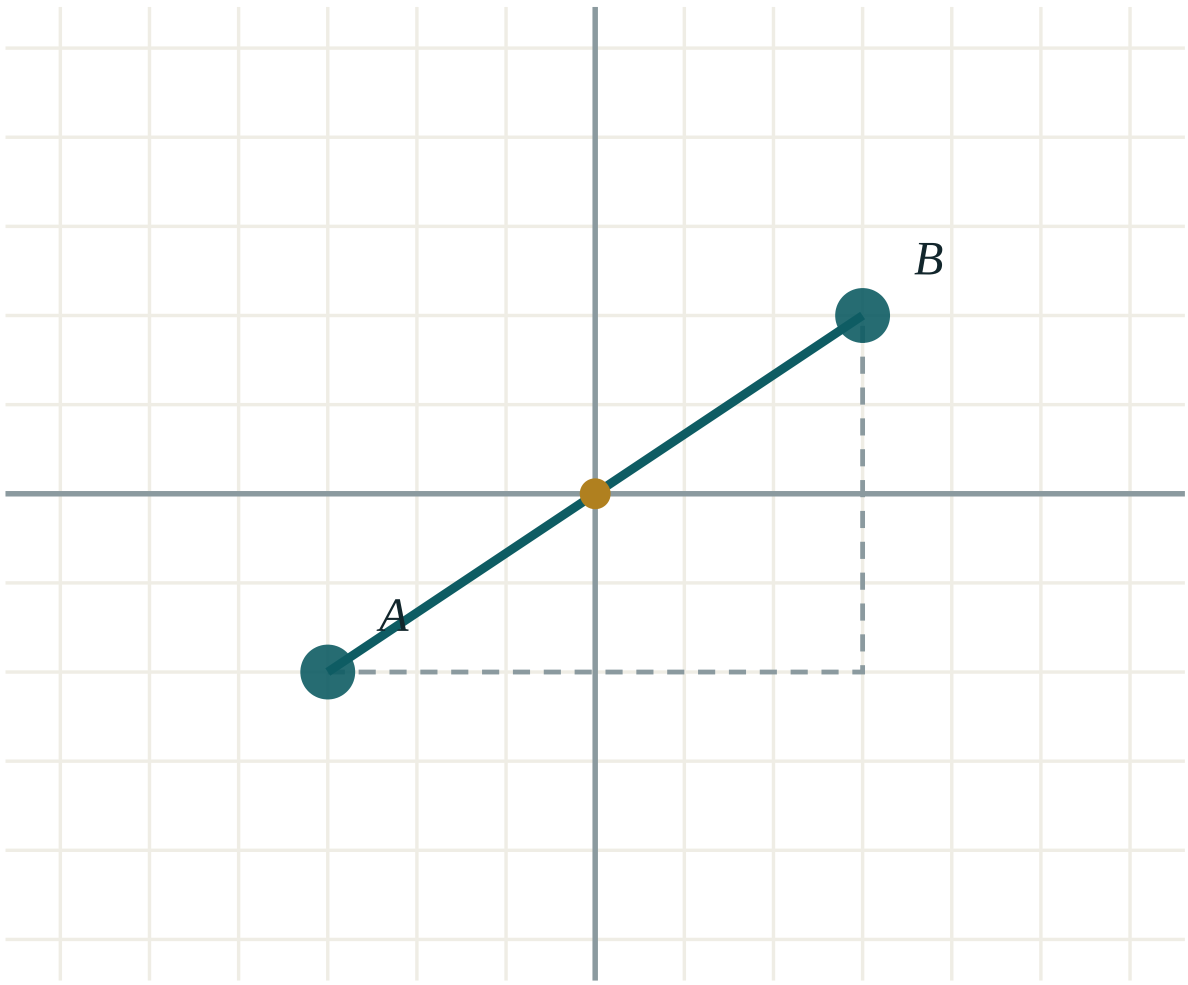
ANSWER

Centre $(1; -4)$, radius 5; the point is on the circle

03 Interactive model

Drag A and B and watch the distance readout follow the two coordinate differences.

Distance between two points



A **(-3, -2)** | B **(3, 2)** | $x_2 - x_1$ **6** | $y_2 - y_1$ **4** | distance AB **7.21** | midpoint M **(0, 0)**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Distance	Masofa	Расстояние
Distance formula	Masofa formulasi	Формула расстояния
Right-angled triangle	To‘g‘ri burchakli uchburchak	Прямоугольный треугольник
Circle	Aylana	Окружность
Centre	Markaz	Центр
Radius	Radius	Радиус
Equation of a circle	Aylana tenglamasi	Уравнение окружности
Locus	Geometrik o‘rin	Геометрическое место точек
Inside / outside	Ichida / tashqarisida	Внутри / снаружи
Unit circle	Birlik aylana	Единичная окружность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The distance between $(0; 0)$ and $(6; 8)$ is:

- 10
- 14
- 48
- 7

The centre of $(x - 2)^2 + (y + 5)^2 = 9$ is:

- $(-2; 5)$
- $(2; -5)$
- $(2; 5)$
- $(-2; -5)$

The radius of $x^2 + y^2 = 49$ is:

- 49
- 7
- 24.5
- $\sqrt{7}$

The distance between $(1; 1)$ and $(4; 5)$ is:

- 5
- 7
- $\sqrt{7}$
- 25

Is $(3; 3)$ inside $x^2 + y^2 = 25$?

- Yes
- No, it is on it
- No, it is outside
- Cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the distance between (0; 0) and (3; 4).*

   ANSWER 5

2. *Find the distance between (1; 0) and (1; 7).*

   ANSWER 7

3. *Find the distance between (2; 3) and (6; 6).*

   ANSWER 5

4. *Write the equation of the circle, centre (0; 0), radius 6.*

   ANSWER $x^2 + y^2 = 36$

5. *State the centre and radius of* $(x - 4)^2 + (y - 1)^2 = 16$.

   ANSWER $(4; 1)$, $r = 4$

6. *State the radius of* $x^2 + y^2 = 100$.

   ANSWER 10

7. *Is (0; 0) inside* $x^2 + y^2 = 9$?

   ANSWER Yes — $0 < 9$

Medium · the standard question

7 PROBLEMS

1. *Find the distance between $(-3; 2)$ and $(1; -1)$.*

   ANSWER 5

2. *Find the distance between $(-2; -5)$ and $(4; 3)$.*

   ANSWER 10

3. *Write the equation of the circle, centre $(3; -2)$, radius 7.*

   ANSWER $(x - 3)^2 + (y + 2)^2 = 49$

4. *State the centre and radius of $(x + 5)^2 + (y - 3)^2 = 36$.*

   ANSWER $(-5; 3)$, $r = 6$

5. *Find the exact distance between $(0; 0)$ and $(2; 3)$.*

   ANSWER $\sqrt{13}$

6. *Does $(5; 12)$ lie on $x^2 + y^2 = 169$?*

   ANSWER Yes — $25 + 144 = 169$

7. *The centre of a circle is $(2; 2)$ and it passes through $(2; 7)$. Find its equation.*

   ANSWER $(x - 2)^2 + (y - 2)^2 = 25$

Hard · stretch and reason

7 PROBLEMS

1. *Show that (1; 2), (4; 6), (8; 3) form an isosceles triangle.*

   ANSWER $AB = 5, BC = 5, AC = \sqrt{50}$ — two sides equal

2. *Show that (0; 0), (4; 0), (4; 3) is right-angled.*

   ANSWER $16 + 9 = 25$ — Pythagoras holds, so the angle at $(4; 0)$ is $90°$

3. *A circle has centre (1; −1) and passes through (4; 3). Find its equation.*

   ANSWER $(x - 1)^2 + (y + 1)^2 = 25$

4. *Find the equation of the circle whose diameter joins (1; 2) and (7; 10).*

   ANSWER centre $(4; 6)$, $r = 5$: $(x - 4)^2 + (y - 6)^2 = 25$

5. *Where does $x^2 + y^2 = 25$ cut the axes?*

   ANSWER $(\pm 5; 0)$ and $(0; \pm 5)$

6. *Is (−3; 4) inside, on or outside $(x + 1)^2 + (y - 1)^2 = 16$?*

   ANSWER $4 + 9 = 13 < 16$ — inside

7. *Find the perimeter of the triangle (0; 0), (6; 0), (6; 8).*

   ANSWER $6 + 8 + 10 = 24$

07 Homework

Homework — 6 problems

Geometry 8, Темы 32–33, pp. 72–74. Leave surds exact unless a decimal is asked for.

1. *Find the distance between (2; 1) and (10; 7).*
2. *Find the distance between (−4; 3) and (2; −5).*
3. *Write the equation of the circle, centre (−1; 4), radius 3.*
4. *State the centre and radius of $(x - 6)^2 + (y + 2)^2 = 81$.*
5. *Show that (0; 0), (5; 0), (0; 12) is right-angled and find its perimeter.*

6. *Is (2; 6) inside, on or outside $x^2 + y^2 = 40$?*

The equation of a straight line

Every straight line is one linear equation — and every linear equation is one straight line.

LESSON 37 | 1 × 40 MIN | GEOMETRY 8, TEMA 34 | STAGE 9 · 10.1–10.2

LESSON CLOCK

5' | 12' | 12' | 7' | 4'

Plot three points and join them	5 min
Gradient and intercept	12 min
Through two points	12 min
Special lines	7 min
Homework	4 min

LEARNING OBJECTIVES

- Write the equation of a line from its gradient and a point.
- Find the equation of a line through two given points.
- Recognise the equations of horizontal and vertical lines.
- Decide whether a point lies on a line.

TEXTBOOK

National	Тема 34, pp. 75–77
Cambridge	Learner’s Book pp. 214–224
Workbook	Workbook 10.1

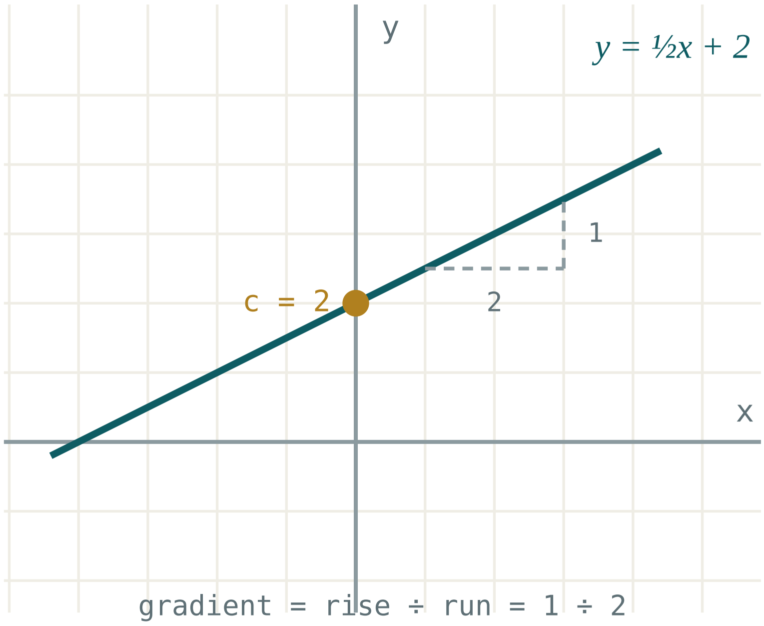
01 Explanation

The two standard forms

EQUATION OF A STRAIGHT LINE

$$y = kx + b \text{ (gradient–intercept) or } Ax + By + C = 0 \text{ (general)}$$

In $y = kx + b$, k is the gradient and b the height at which the line crosses the y-axis.



The gradient is rise ÷ run; b is where the line meets the y-axis.

A point lies on a line exactly when its coordinates **satisfy** the equation. Does $(3; 7)$ lie on $y = 2x + 1$? $2 \cdot 3 + 1 = 7$ ✓ — yes.

Through two points

Two points fix a line completely. Find the gradient first, then the intercept:

$$k = \frac{y_2 - y_1}{x_2 - x_1}$$

Through $(2; 3)$ and $(6; 11)$: $k = \frac{8}{4} = 2$, so $y = 2x + b$. Substituting $(2; 3)$ gives $3 = 4 + b$, so $b = -1$ and the line is $y = 2x - 1$. Check with the second point: $12 - 1 = 11$ ✓.

TWO SPECIAL CASES

A **horizontal** line is $y = c$ — gradient 0. A **vertical** line is $x = c$ — it has no gradient at all, and it cannot be written as $y = kx + b$, because the run is zero.

02 Worked examples

EXAMPLE 1 **Find the equation of the line through $A(1; 4)$ and $B(5; 12)$.**

1. $k = \dfrac{12 - 4}{5 - 1} = 2$

   Same order top and bottom.

2. $y = 2x + b$

3. $4 = 2 \cdot 1 + b$

   Substitute A.

4. $b = 2$

5. $y = 2x + 2$

   Check B: $10 + 2 = 12$ ✓

ANSWER

$y = 2x + 2$

EXAMPLE 2 **Where do $y = 3x - 2$ and $y = x + 6$ meet?**

1. $3x - 2 = x + 6$

   At the meeting point both give the same y.

2. $2x = 8, x = 4$

3. $y = 4 + 6 = 10$

   Substitute into either equation.

4. $(4; 10)$

   Check in the other: $12 - 2 = 10$ ✓

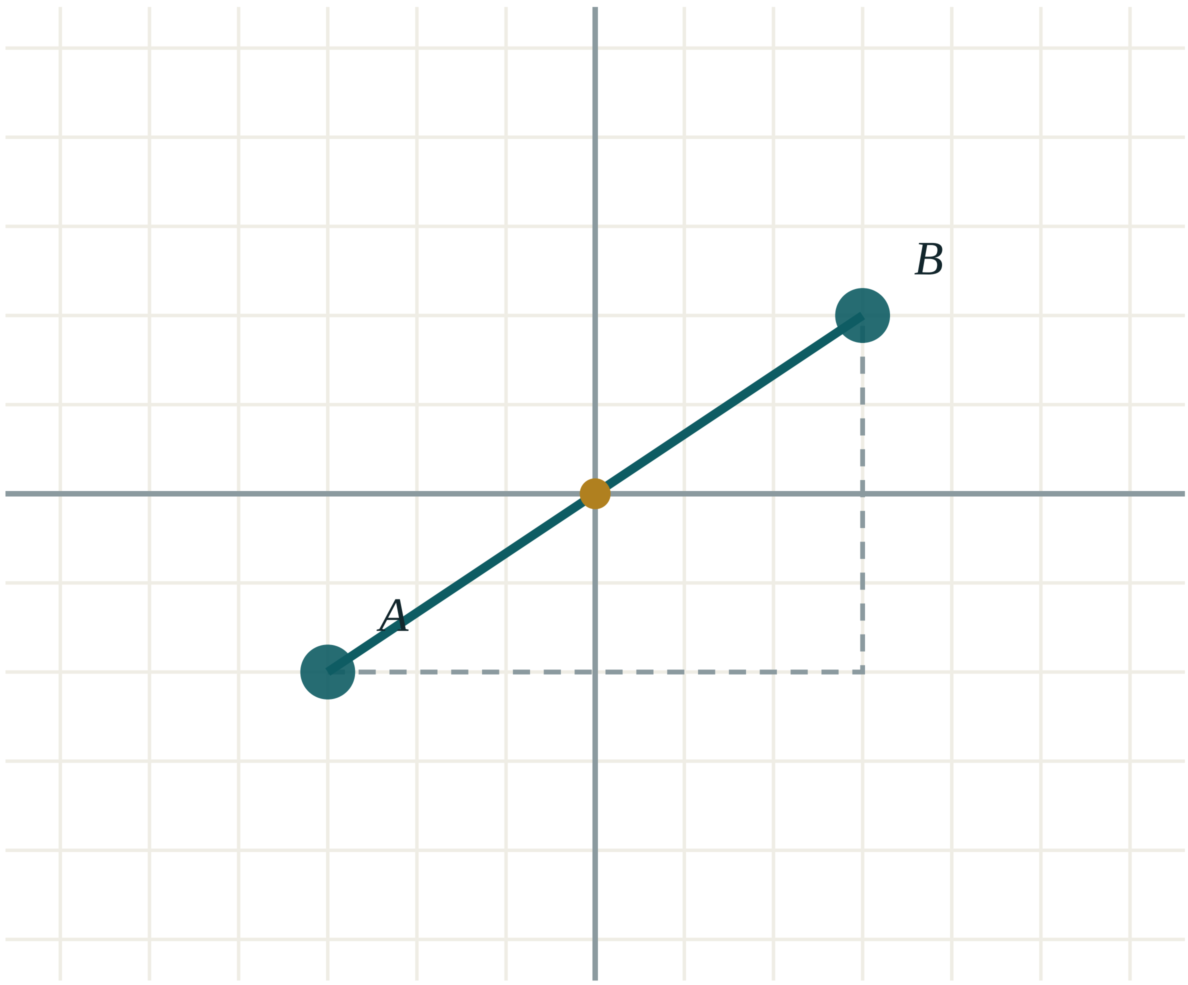
ANSWER

$(4; 10)$

03 Interactive model

Drag the two points and read the gradient from the change in x and the change in y.

A line through two points



A (-3, -2) | B (3, 2) | $x_2 - x_1$ 6 | $y_2 - y_1$ 4 | distance AB 7.21 | midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Straight line	To‘g‘ri chiziq	Прямая
Equation of a line	To‘g‘ri chiziq tenglamasi	Уравнение прямой
Gradient	Burchak koeffitsienti	Угловой коэффициент
Intercept	Kesma (o‘qdagi)	Отрезок на оси
General form	Umumiy ko‘rinish	Общий вид
Horizontal line	Gorizontal chiziq	Горизонтальная прямая
Vertical line	Vertikal chiziq	Вертикальная прямая
Satisfies the equation	Tenglamani qanoatlantiradi	Удовлетворяет уравнению
Intersection point	Kesishish nuqtasi	Точка пересечения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The gradient of $y = 4 - 2x$ is:

- 4
- 2
- -2
- -4

The line through $(0; 1)$ and $(3; 7)$ is:

- $y = 2x + 1$
- $y = 3x + 1$
- $y = 2x$
- $y = x + 1$

A horizontal line has equation:

- $x = c$
- $y = c$
- $y = x$
- $x + y = 0$

Does $(2; 5)$ lie on $y = 3x - 1$?

- Yes
- No
- Only if x > 0
- Cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *State the gradient and intercept of $y = 3x + 5$.*

   ANSWER $k = 3, b = 5$

2. *Does $(1; 4)$ lie on $y = 4x$?*

   ANSWER Yes.

3. *Write the equation of the horizontal line through $(2; 7)$.*

   ANSWER $y = 7$

4. *Write the equation of the vertical line through $(2; 7)$.*

   ANSWER $x = 2$

5. *Find the gradient through $(0; 0)$ and $(3; 9)$.*

   ANSWER 3

6. *Where does $y = 2x - 6$ cut the y-axis?*

   ANSWER $(0; -6)$

7. *Where does $y = 2x - 6$ cut the x-axis?*

   ANSWER $(3; 0)$

Medium · the standard question

7 PROBLEMS

1. *Find the equation through $(0; 3)$ with gradient -2.*

   ANSWER $y = -2x + 3$

2. *Find the equation through $(1; 5)$ and $(3; 11)$.*

   ANSWER $y = 3x + 2$

3. *Find the equation through $(-2; 1)$ and $(2; 9)$.*

   ANSWER $y = 2x + 5$

4. *Rearrange $2x + y - 8 = 0$ into $y = kx + b$.*

   ANSWER $y = -2x + 8$

5. *Where do $y = x + 1$ and $y = 5 - x$ meet?*

   ANSWER $(2; 3)$

6. *Find the equation through $(4; 0)$ and $(0; 8)$.*

   ANSWER $y = -2x + 8$

7. *Does $(-1; -5)$ lie on $y = 4x - 1$?*

   ANSWER Yes — $-4 - 1 = -5$

Hard · stretch and reason

7 PROBLEMS

1. *Find the equation through $(2; -3)$ and $(-4; 9)$.*

   ANSWER $y = -2x + 1$

2. *Where do $2x + y = 7$ and $x - y = 2$ meet?*

   ANSWER $(3; 1)$

3. *Show that $(1; 3)$, $(4; 9)$ and $(6; 13)$ are collinear.*

   ANSWER All satisfy $y = 2x + 1$.

4. *A line has gradient 3 and passes through $(2; 1)$. Find where it cuts each axis.*

   ANSWER $y = 3x - 5$: $(0; -5)$ and $(\frac{5}{3}; 0)$

5. *Find the equation of the line through $(0; 0)$ and the midpoint of $(2; 4)$ and $(6; 8)$.*

   ANSWER midpoint $(4; 6)$, so $y = \frac{3}{2}x$

6. *The line $y = kx + 2$ passes through $(4; -6)$. Find k.*

   ANSWER $k = -2$

7. *Find the equation of the perpendicular bisector of the segment from $(0; 0)$ to $(4; 2)$.*

   ANSWER midpoint $(2; 1)$, gradient -2: $y = -2x + 5$

07 Homework

Homework — 5 problems

Geometry 8, Тема 34, pp. 75–77. Always check your equation with the second point.

1. *Find the equation through $(0; -2)$ with gradient 5.*
2. *Find the equation through $(1; 2)$ and $(4; 11)$.*
3. *Where do $y = 3x - 4$ and $y = x + 2$ meet?*
4. *Write the horizontal and the vertical line through $(-3; 6)$.*

5. *Show that (2; 5), (4; 9) and (7; 15) are collinear.*

The coordinate method in practice

Choose your axes well and a hard proof becomes two lines of arithmetic.

LESSON 38 | 1 × 40 MIN | GEOMETRY 8, TEMA 34 | STAGE 9 · 3.1

LESSON CLOCK

5' | 12' | 12' | 7' | 4'

Place a rectangle on axes three ways	5 min
Proving with coordinates	12 min
Classifying triangles	12 min
Finding a missing vertex	7 min
Homework	4 min

LEARNING OBJECTIVES

- Place a figure on axes so that its coordinates are as simple as possible.
- Prove that a quadrilateral is a parallelogram, rectangle or rhombus using coordinates.
- Classify a triangle from the lengths of its sides.
- Find an unknown vertex from a condition.

TEXTBOOK

National	Тема 34, pp. 75–77
Cambridge	Learner’s Book pp. 56–60
Workbook	Workbook 3.1

01 Explanation

Choose the axes yourself

When a problem gives you a shape but no coordinates, **you** choose where to put it. A good choice puts one vertex at the origin and one side along an axis, so that as many coordinates as possible are zero.

SHAPE	A GOOD PLACING
Rectangle	$(0;0), (a;0), (a;b), (0;b)$
Right triangle	$(0;0), (a;0), (0;b)$
Isosceles triangle	$(-a;0), (a;0), (0;h)$
Parallelogram	$(0;0), (a;0), (a+b;c), (b;c)$

The letters keep the proof general — it then holds for **every** rectangle, not just the one you drew.

The three tools

EVERYTHING COMES FROM THESE

1. **Distance** $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ — for equal sides, and for Pythagoras.
2. **Midpoint** $\left(\frac{x_1 + x_2}{2};\ \frac{y_1 + y_2}{2}\right)$ — for bisection and for medians.
3. **Gradient** $\frac{y_2 - y_1}{x_2 - x_1}$ — for parallel (equal) and perpendicular (product -1).

Which shape you are proving decides which tool you reach for:

TO PROVE	SHOW THAT
parallelogram	the diagonals share a midpoint
rectangle	a parallelogram whose diagonals are also equal
rhombus	all four sides equal
right-angled triangle	Pythagoras holds, or two gradients multiply to -1
isosceles triangle	two side lengths are equal

02 Worked examples

EXAMPLE 1 **Prove that $A(0; 0)$, $B(4; 0)$, $C(4; 3)$, $D(0; 3)$ is a rectangle.**

1. Midpoint of AC: $(2; 1.5)$

2. Midpoint of BD: $(2; 1.5)$

   Diagonals bisect each other ⟹ parallelogram.

3. $AC = \sqrt{16 + 9} = 5$

4. $BD = \sqrt{16 + 9} = 5$

   The diagonals are also equal.

5. A parallelogram with equal diagonals is a rectangle.

   Theorem from Quarter I.

ANSWER

A rectangle

EXAMPLE 2 **Classify the triangle $A(0; 0)$, $B(6; 0)$, $C(3; 4)$.**

1. $AB = 6$

   Along the axis.

2. $AC = \sqrt{9 + 16} = 5$

3. $BC = \sqrt{9 + 16} = 5$

   Two sides equal.

4. $5^2 + 5^2 = 50 \neq 36$

   Not right-angled.

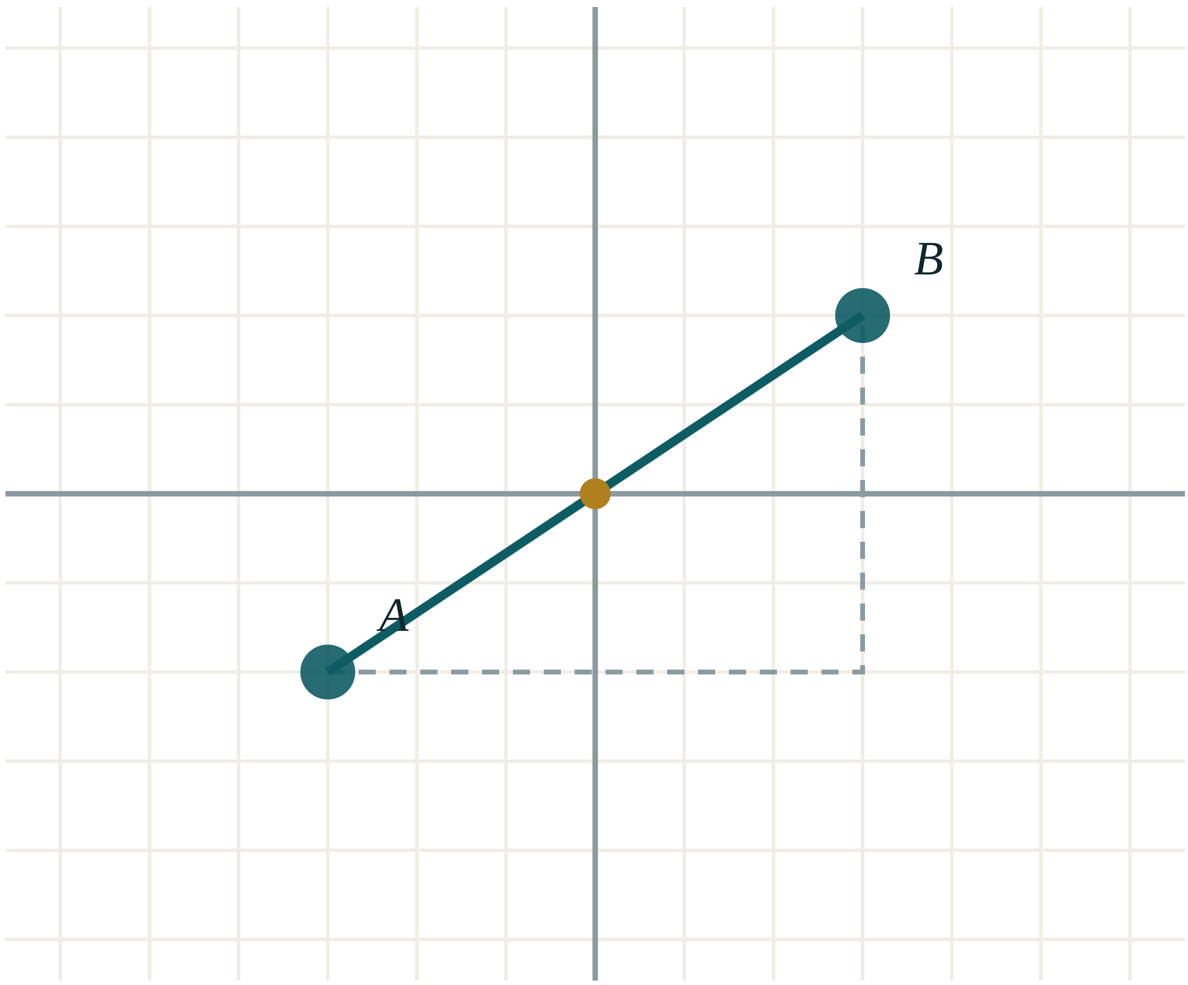
ANSWER

Isosceles, but not right-angled

03 Interactive model

Place the vertices with the model, then read off the distances the proof needs.

Measure a side of your figure



A (-3, -2) | B (3, 2) | $x_2 - x_1$ 6 | $y_2 - y_1$ 4 | distance AB 7.21 | midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Coordinate method	Koordinatalar usuli	Метод координат
To place on axes	O‘qlarga joylashtirish	Расположить на осях
Prove	Isbotlash	Доказать
Isosceles	Teng yonli	Равнобедренный
Scalene	Turli tomonli	Разносторонний
Collinear	Bir to‘g‘ri chiziqda yotuvchi	Лежащие на одной прямой
Vertex	Uchi	Вершина
Diagonal	Diagonal	Диагональ
Converse	Teskari teorema	Обратная теорема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

To prove a quadrilateral is a parallelogram with coordinates, show:

- all sides equal
- the diagonals share a midpoint
- one angle is 90°
- the diagonals are equal

$(0; 0), (5; 0), (5; 5), (0; 5)$ is a:

- rectangle only
- rhombus only
- square
- trapezium

A triangle with sides $5, 5, 6$ is:

- equilateral
- isosceles
- scalene
- right-angled

Two lines are perpendicular when their gradients:

- are equal
- multiply to -1
- add to 0
- are both positive

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the length of the side from (0; 0) to (0; 7).*

   ANSWER 7

2. *Is (0;0), (4;0), (4;4), (0;4) a square?*

   ANSWER Yes.

3. *Find the midpoint of the diagonal from (0; 0) to (6; 8).*

   ANSWER $(3; 4)$

4. *Classify the triangle (0;0), (3;0), (0;4).*

   ANSWER Right-angled, scalene.

5. *Find the gradient from (0; 0) to (2; 6).*

   ANSWER 3

6. *Are the segments from (0;0) to (2;1) and (0;0) to (−1;2) perpendicular?*

   ANSWER Yes — $\frac{1}{2} \cdot (-2) = -1$

7. *Find the perimeter of (0;0), (3;0), (3;4).*

   ANSWER 12

Medium · the standard question

7 PROBLEMS

1. *Prove that (1;1), (5;1), (5;4), (1;4) is a rectangle.*

   ANSWER Diagonals both have midpoint $(3; 2.5)$ and both equal 5.

2. *Classify the triangle (0;0), (4;0), (2;5).*

   ANSWER Isosceles — $AC = BC = \sqrt{29}$

3. *Show that (0;0), (4;3), (8;0), (4;−3) is a rhombus.*

   ANSWER All four sides equal 5.

4. *Find the fourth vertex of a parallelogram with (0;0), (5;1), (7;5).*

   ANSWER $(2; 4)$

5. *Show that the triangle (0;0), (6;0), (0;8) is right-angled and find its area.*

   ANSWER $36 + 64 = 100$ ✓; area 24

6. *Find the length of the median from (0;0) to the side joining (6;0) and (2;4).*

   ANSWER midpoint $(4; 2)$, length $\sqrt{20} = 2\sqrt{5}$

7. *Are (1;2), (3;6) and (6;12) collinear?*

   ANSWER Yes — both gradients equal 2, and all three satisfy $y = 2x$.

Hard · stretch and reason

7 PROBLEMS

1. *Prove that the diagonals of the rectangle (0;0), (a;0), (a;b), (0;b) are equal, for all a and b.*

   ANSWER Both diagonals equal $\sqrt{a^2 + b^2}$.

2. *Prove that the midpoints of the sides of any quadrilateral form a parallelogram.*

   ANSWER With vertices $(x_1;y_1)\ldots(x_4;y_4)$, both diagonals of the midpoint figure have midpoint $\left(\frac{x_1+x_2+x_3+x_4}{4}; \frac{y_1+y_2+y_3+y_4}{4}\right)$.

3. *Show that the triangle (1;1), (4;5), (8;2) is right-angled.*

   ANSWER $25 + 25 = 50$ ✓ — the right angle is at $(4; 5)$

4. *Find the point on the x-axis equidistant from (1; 2) and (5; 4).*

   ANSWER $(4.5; 0)$

5. *Prove that the medians of the triangle (0;0), (6;0), (0;6) meet at (2; 2).*

   ANSWER Each median passes through $(2; 2)$ — the centroid is the average of the vertices.

6. *The vertices of a square are (0;0), (4;2) and (2;6). Find the fourth.*

   ANSWER $(-2; 4)$

7. *Show that (0;0), (6;0), (8;3), (2;3) is a parallelogram but not a rhombus.*

   ANSWER Both diagonals have midpoint $(4; 1.5)$, so it is a parallelogram; but the sides are 6 and $\sqrt{13}$, which are unequal.

07 Homework

Homework — 5 problems

Geometry 8, Тема 34. State which of the three tools you used at each step.

1. *Prove that (0;0), (6;0), (6;4), (0;4) is a rectangle.*
2. *Classify the triangle (0;0), (8;0), (4;3).*
3. *Show that (0;0), (3;4), (8;4), (5;0) is a rhombus.*

4. *Find the fourth vertex of a parallelogram with (1;1), (6;2), (8;7).*
5. *Show that (2;1), (5;7) and (8;13) are collinear.*

Home / Grade 8 / Geometry / Quarter 3

The concept of a vector

A quantity that needs a direction as well as a size — and the notation that carries both.

LESSON 39 | 1 × 40 MIN | GEOMETRY 8, TEMA 35 | STAGE 9 · 13.1

LESSON CLOCK

5' | 12' | 12' | 7' | 4'

Which of these need a direction?	5 min
Vectors and scalars	12 min
Length and notation	12 min
Equal and opposite	7 min
Homework	4 min

LEARNING OBJECTIVES

- Distinguish a vector from a scalar.
- Use the notations AB with an arrow, and a in bold.
- Find the length (modulus) of a vector.
- Recognise equal, opposite and collinear vectors.

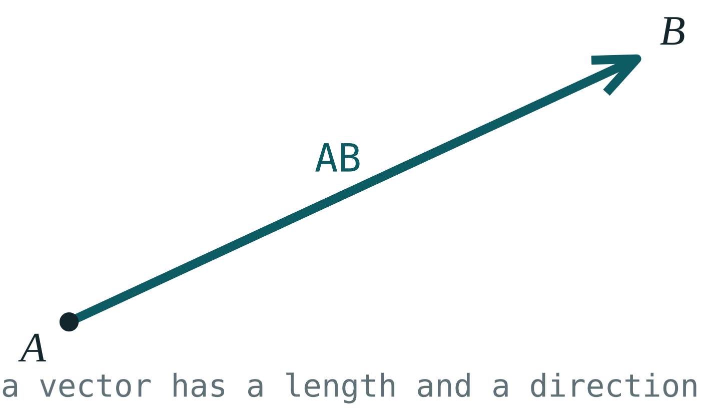
TEXTBOOK

National	Тема 35, pp. 78–80
Cambridge	Learner's Book pp. 284–290
Workbook	Workbook 13.1

01 Explanation

Size is not always enough

"The bus travelled 40 km" is a **scalar** — one number. "The bus travelled 40 km north" is a **vector**: a size *and* a direction. Force, velocity and displacement are vectors; mass, time, temperature and area are scalars.



A vector is a directed segment: it has a length and a direction, but no fixed position.

A vector is drawn as an arrow from its **initial point** to its **terminal point** and written AB with an arrow over it, or as a single bold letter $\boldsymbol{a}$.

LENGTH OF A VECTOR

If $\boldsymbol{a} = (a_1; a_2)$ then

$$|\,\boldsymbol{a}\,| = \sqrt{a_1^2 + a_2^2}$$

— the distance formula again. A length is a scalar, and it is never negative.

Equal, opposite, collinear

RELATIONSHIP	MEANS	IN COORDINATES
equal	same length, same direction	$a_1 = b_1$ and $a_2 = b_2$
opposite	same length, reversed direction	$(-a_1; -a_2)$
collinear	parallel — same or opposite direction	one is a multiple of the other
zero vector	no length, no direction	$(0; 0)$

A VECTOR HAS NO ADDRESS

Two arrows of the same length pointing the same way are the **same vector**, even in different corners of the page. That freedom is exactly what makes vectors useful — you may slide one anywhere to add it to another.

02 Worked examples

EXAMPLE 1 **Find the vector AB and its length, for $A(1; 2)$ and $B(5; 5)$.**

1. $AB = (5 - 1; 5 - 2)$

   End minus start — always in that order.

2. $AB = (4; 3)$

3. $|AB| = \sqrt{16 + 9} = 5$

ANSWER

$(4; 3)$, length 5

EXAMPLE 2 **Are $\boldsymbol{a} = (2; -3)$ and $\boldsymbol{b} = (-6; 9)$ collinear?**

1. $-6 = -3 \cdot 2$

   Compare the first coordinates.

2. $9 = -3 \cdot (-3)$

   The same multiplier works for the second.

3. $\boldsymbol{b} = -3\boldsymbol{a}$

   One is a multiple of the other.

4. They are collinear, pointing in opposite directions.

   The multiplier is negative.

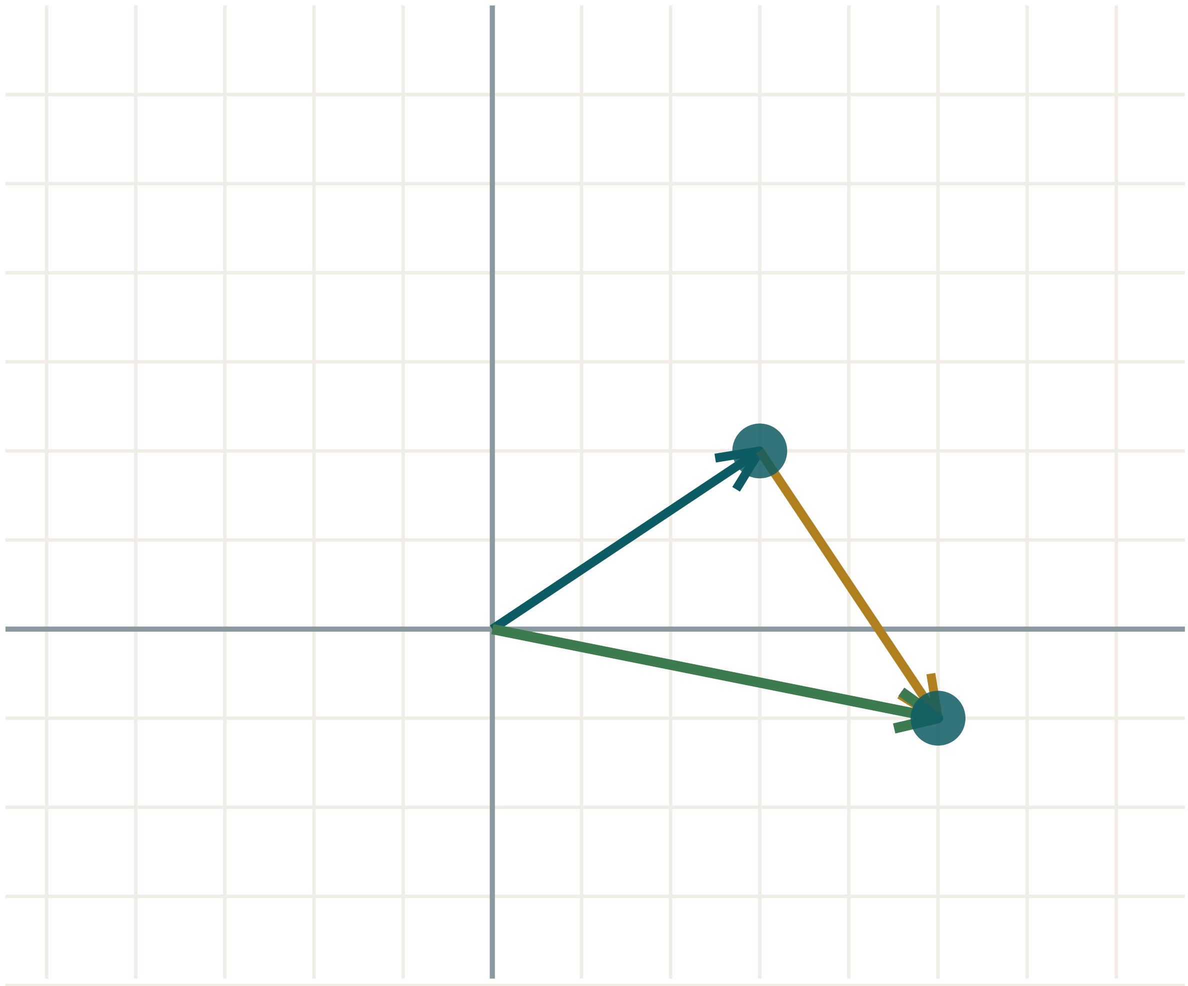
ANSWER

Yes — $\boldsymbol{b} = -3\boldsymbol{a}$

03 Interactive model

Drag the tip of the arrow and watch the coordinates and the length change together.

A vector and its length



a (3; 2)	b (2; -3)	a + b (5; -1)	| a | 3.61	| b | 3.61	| a + b | 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Vector	Vektor	Вектор
Scalar	Skalyar	Скаляр
Directed segment	Yo‘naltirilgan kesma	Направленный отрезок
Initial point	Boshi	Начало вектора
Terminal point	Oxiri	Конец вектора
Modulus (length)	Vektor uzunligi (moduli)	Модуль (длина) вектора
Equal vectors	Teng vektorlar	Равные векторы
Opposite vector	Qarama-qarshi vektor	Противоположный вектор
Zero vector	Nol vektor	Нулевой вектор
Collinear vectors	Kollinear vektorlar	Коллинеарные векторы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which of these is a vector?

- mass
- temperature
- velocity
- area

For $A(2; 1)$ and $B(6; 4)$, AB is:

- $(4; 3)$
- $(-4; -3)$
- $(8; 5)$
- $(3; 4)$

The length of $(-3; 4)$ is:

- 1
- 5
- 7
- -5

The vector opposite to $(5; -2)$ is:

- $(-5; 2)$
- $(2; -5)$
- $(5; 2)$
- $(-5; -2)$

$(1; 2)$ and $(3; 6)$ are:

- equal
- opposite
- collinear
- perpendicular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Find AB for $A(0; 0)$, $B(3; 4)$.*

   ANSWER $(3; 4)$

2. *Find the length of $(6; 8)$.*

   ANSWER 10

3. *Find the length of $(0; -7)$.*

   ANSWER 7

4. *Write the vector opposite to $(2; 5)$.*

   ANSWER $(-2; -5)$

5. *Is force a vector or a scalar?*

   ANSWER A vector.

6. *Find AB for $A(1; 1)$, $B(1; 6)$.*

   ANSWER $(0; 5)$

7. *Find the length of $(-1; 0)$.*

   ANSWER 1

Medium · the standard question

7 PROBLEMS

1. *Find AB and $|AB|$ for $A(-2; 3)$, $B(4; -5)$.*

   ANSWER $(6; -8)$, length 10

2. *Are $(4; 6)$ and $(6; 9)$ collinear?*

   ANSWER Yes — $(6; 9) = 1.5(4; 6)$

3. *Are $(2; 3)$ and $(3; 2)$ collinear?*

   ANSWER No — no single multiplier works.

4. *Find the length of $(-5; 12)$.*

   ANSWER 13

5. *$B(4; 7)$ and $AB = (1; 3)$. Find A.*

   ANSWER $A(3; 4)$

6. *Find the exact length of $(2; 3)$.*

   ANSWER $\sqrt{13}$

7. *Write a vector equal to AB where $A(0; 0)$, $B(2; -1)$, starting at $(5; 5)$.*

   ANSWER From $(5; 5)$ to $(7; 4)$

Hard · stretch and reason

7 PROBLEMS

1. $|a| = 5$ and $a = (3; k)$. Find all k.

   ANSWER $k = \pm 4$

2. Find a vector of length 10 collinear with $(3; 4)$.

   ANSWER $(6; 8)$ or $(-6; -8)$

3. Show that $A(1;1)$, $B(4;5)$, $C(7;9)$ are collinear using vectors.

   ANSWER $AB = (3; 4)$, $BC = (3; 4)$ — equal vectors, so the three points lie on one line.

4. $|a| = 13$ and $a = (k; 12)$. Find all k.

   ANSWER $k = \pm 5$

5. Find a unit vector (length 1) in the direction of $(6; 8)$.

   ANSWER $(0.6; 0.8)$

6. $A(2; 1)$, $B(6; 4)$, $C(9; 8)$. Is ABC a straight line?

   ANSWER No — $AB = (4; 3)$ and $BC = (3; 4)$ are not multiples of each other.

7. Explain why a vector has no fixed position but a directed segment does.

   ANSWER A vector records only length and direction; any two arrows agreeing on both represent the same vector, while a directed segment also carries its start point.

07 Homework

Homework — 5 problems

Geometry 8, Тема 35, pp. 78–80. Draw every vector as an arrow on squared paper.

1. Find AB and $|AB|$ for $A(1; 2)$, $B(7; 10)$.
2. Find the length of $(-8; 15)$.
3. Write the vector opposite to $(-4; 9)$.
4. Are $(2; -6)$ and $(-1; 3)$ collinear? Justify.

5. Name three vector quantities and three scalar quantities from physics.

Adding and subtracting vectors

Nose to tail, or by the parallelogram — two pictures of one operation.

LESSON 40–41 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 36–37 | STAGE 9 · 13.1–13.2

LESSON CLOCK

12' | 28' | 24' | 8' | 8'

Two walks, one displacement	12 min
The triangle and parallelogram rules	28 min
Subtraction	24 min
In coordinates	8 min
Homework	8 min

LEARNING OBJECTIVES

- Add two vectors by the triangle rule and by the parallelogram rule.
- Subtract a vector by adding its opposite.
- Use the relation $AB + BC = AC$.
- Add and subtract in coordinates.

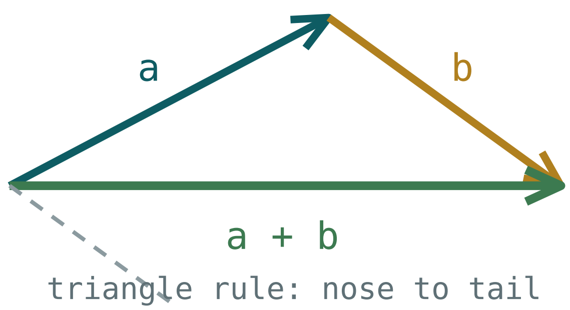
TEXTBOOK

National	Темы 36–37, pp. 81–84
Cambridge	Learner's Book pp. 284–296
Workbook	Workbook 13.1–13.2

01 Explanation

Adding: nose to tail

Walk 3 km east, then 4 km north. Your displacement is not 7 km — it is 5 km north-east. That is vector addition, and the picture is the whole rule:



The triangle rule: place the second vector at the tip of the first, and the sum runs from the first tail to the last tip.

TRIANGLE RULE

$$AB + BC = AC$$

Whatever route you take, only the start and the end matter.

The **parallelogram rule** gives the same answer with both vectors drawn from a common point: complete the parallelogram, and the sum is the diagonal from that point.

In coordinates, add componentwise:

$$(a_1; a_2) + (b_1; b_2) = (a_1 + b_1; a_2 + b_2)$$

Addition is **commutative — $\boldsymbol{a} + \boldsymbol{b} = \boldsymbol{b} + \boldsymbol{a}$ —** which is exactly why the parallelogram closes.

Subtracting

To subtract, add the opposite:

$$\boldsymbol{a} - \boldsymbol{b} = \boldsymbol{a} + (-\boldsymbol{b})$$

In coordinates that is simply $(a_1 - b_1; a_2 - b_2)$. In a picture, $\boldsymbol{a} - \boldsymbol{b}$ is the vector **from the tip of $\boldsymbol{b}$ to the tip of $\boldsymbol{a}$** when both are drawn from the same point — the other diagonal of the parallelogram.

THE RELATION WORTH MEMORISING

$$AB = OB - OA$$

Any vector is the position of its end minus the position of its start. That single line turns most vector problems into subtraction.

LENGTHS DO NOT ADD

$|\boldsymbol{a} + \boldsymbol{b}|$ is **not** $|\boldsymbol{a}| + |\boldsymbol{b}|$ unless the two point the same way. For $(3; 0)$ and $(0; 4)$ the lengths are 3 and 4, but the sum has length 5.

02 Worked examples

EXAMPLE 1 $\boldsymbol{a} = (3; -1), \boldsymbol{b} = (-5; 4)$. **Find** $\boldsymbol{a} + \boldsymbol{b}, \boldsymbol{a} - \boldsymbol{b}$ **and** $|\,\boldsymbol{a} + \boldsymbol{b}\,|$.

1. $\boldsymbol{a} + \boldsymbol{b} = (3 - 5;\ -1 + 4) = (-2;\ 3)$

   Add componentwise.

2. $\boldsymbol{a} - \boldsymbol{b} = (3 + 5;\ -1 - 4) = (8;\ -5)$

   Subtract componentwise.

3. $|\,(-2;\ 3)\,| = \sqrt{4 + 9} = \sqrt{13}$

   Leave it exact.

ANSWER

$(-2;\ 3),\ (8;\ -5),\ \sqrt{13}$

EXAMPLE 2 **In a parallelogram** $ABCD$**, express** AC **and** BD **in terms of** $AB = \boldsymbol{a}$ **and** $AD = \boldsymbol{b}$.

1. $AC = AB + BC$

   Triangle rule, going round through B.

2. $BC = AD = \boldsymbol{b}$

   Opposite sides of a parallelogram are equal vectors.

3. $AC = \boldsymbol{a} + \boldsymbol{b}$

   One diagonal is the sum.

4. $BD = BA + AD = -\boldsymbol{a} + \boldsymbol{b}$

   The other diagonal is the difference.

ANSWER

$AC = \boldsymbol{a} + \boldsymbol{b},\ BD = \boldsymbol{b} - \boldsymbol{a}$

EXAMPLE 3 **A boat heads north at *8 km/h* across a river flowing east at *6 km/h*. Find its actual speed.**

1. $(0; 8) + (6; 0) = (6; 8)$

   The two velocities add as vectors.

2. $|\,(6; 8)\,| = \sqrt{36 + 64}$

3. $= 10$ km/h

   Not 14 — the directions differ.

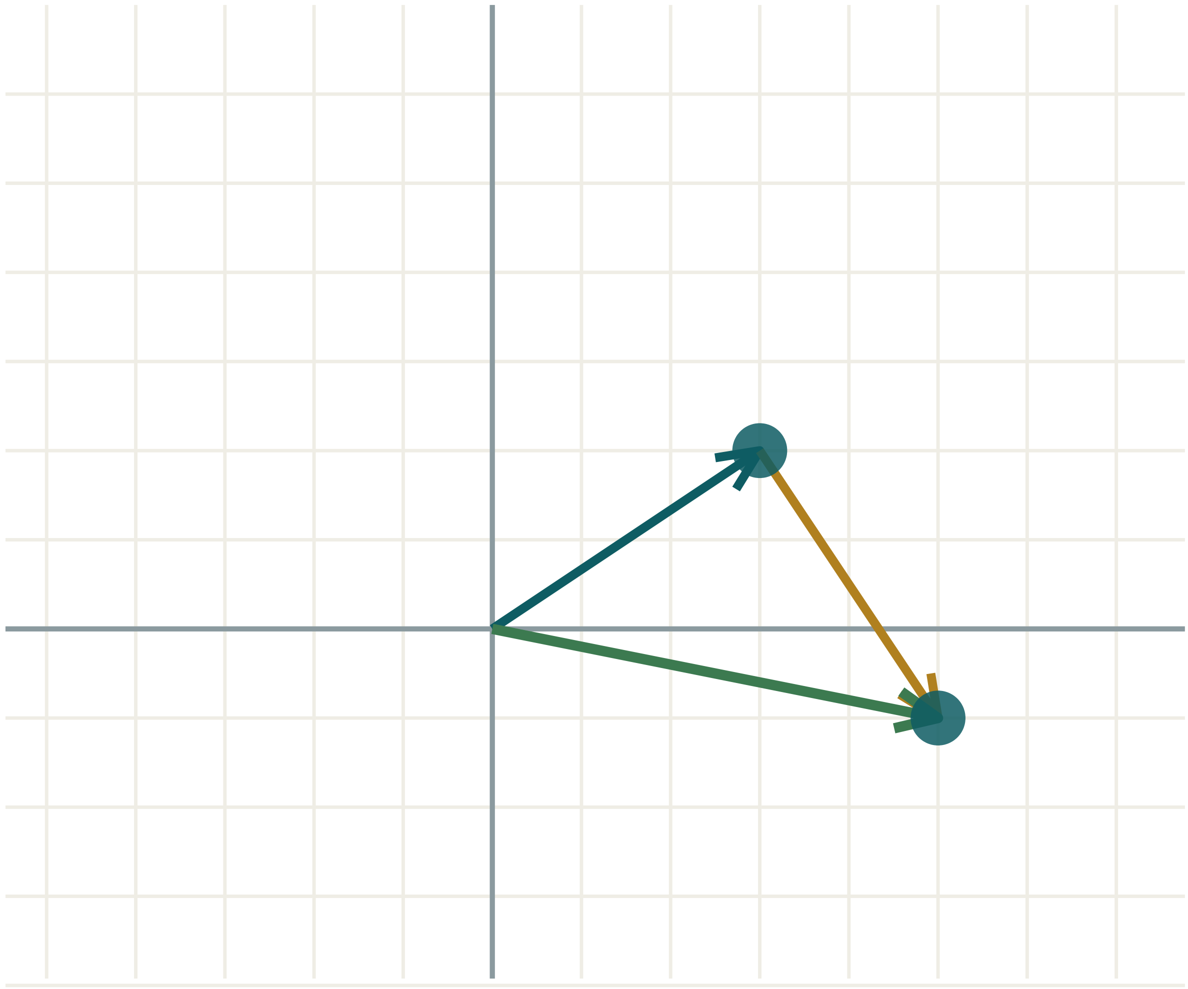
ANSWER

10 km/h

03 Interactive model

Drag the tips of a and b; the green arrow is always their sum, and it closes the triangle.

The triangle rule in action



a **(3; 2)**	b **(2; -3)**	a + b **(5; -1)**	| a | **3.61**	| b | **3.61**	| a + b | **5.1**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sum of vectors	Vektorlar yig‘indisi	Сумма векторов
Difference of vectors	Vektorlar ayirmasi	Разность векторов
Triangle rule	Uchburchak qoidasi	Правило треугольника
Parallelogram rule	Parallelogramm qoidasi	Правило параллелограмма
Resultant	Natijaviy vektor	Равнодействующий вектор
Nose to tail	Uchi-boshiga	Конец к началу
Commutative	O‘rin almashinuvchi	Коммутативный
Associative	Guruhlanuvchi	Ассоциативный
Closed polygon	Yopiq ko‘pburchak	Замкнутый многоугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(2; 5) + (3; -1)$ is:

- $(5; 4)$
- $(5; 6)$
- $(-1; 6)$
- $(6; -5)$

$(4; 1) - (6; 5)$ is:

- $(10; 6)$
- $(-2; -4)$
- $(2; 4)$
- $(-2; 4)$

$AB + BC$ equals:

- AC
- CA
- BA
- 0

If $|\,\boldsymbol{a}\,| = 3$ and $|\,\boldsymbol{b}\,| = 4$ and they are perpendicular, $|\,\boldsymbol{a} + \boldsymbol{b}\,|$ is:

- 7
- 5
- 1
- 12

AB in terms of position vectors is:

- $OA + OB$
- $OB - OA$
- $OA - OB$
- $OA \cdot OB$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Find $(1; 2) + (3; 4)$.

ANSWER $(4; 6)$

2. Find $(5; 7) - (2; 3)$.

ANSWER $(3; 4)$

3. Find $(0; 0) + (-2; 6)$.

ANSWER $(-2; 6)$

4. Find $(4; -1) + (-4; 1)$.

ANSWER $(0; 0)$

5. Simplify $AB + BC$.

ANSWER AC

6. Find $|(3; 4) + (0; 0)|$.

ANSWER 5

7. Find $(7; 2) - (7; 2)$.

ANSWER $(0; 0)$

Medium · the standard question

7 PROBLEMS

1. $a = (2; -3)$, $b = (-1; 5)$. Find $a + b$ and $a - b$.

ANSWER $(1; 2)$ and $(3; -8)$

2. Find $|(3; 4) + (4; -3)|$.

ANSWER $|(7; 1)| = \sqrt{50} = 5\sqrt{2}$

3. Simplify $AB + BC + CD$.

ANSWER AD

4. Simplify $AB + BA$.

ANSWER 0 — the zero vector

5. $OA = (1; 2)$, $OB = (5; 9)$. Find AB.

ANSWER $(4; 7)$

6. A plane flies north at 200 km/h in a wind blowing east at 150 km/h. Find its actual speed.

ANSWER 250 km/h

7. $a = (6; 8)$. Find a vector b with $a + b = (0; 0)$.

ANSWER $(-6; -8)$

Hard · stretch and reason

7 PROBLEMS

1. *In parallelogram ABCD, AB = a and AD = b. Express DB and CA.*

   ANSWER $DB = \boldsymbol{a} - \boldsymbol{b}$, $CA = -\boldsymbol{a} - \boldsymbol{b}$

2. *Simplify AB + BC + CD + DA.*

   ANSWER 0 — the walk returns to its start

3. $|a| = 5$, $|b| = 12$, *and* $a \perp b$. *Find* $|a + b|$ *and* $|a - b|$.

   ANSWER Both 13 — the diagonals of a rectangle are equal.

4. $a = (3; 1)$, $b = (-1; 4)$. *Find* $|a + b|$ *and* $|a| + |b|$, *and compare.*

   ANSWER $|(2; 5)| = \sqrt{29} \approx 5.39$ against $\sqrt{10} + \sqrt{17} \approx 7.28$ — the sum is shorter

5. *M is the midpoint of AB. Show that OM = ½(OA + OB).*

   ANSWER $OM = OA + \frac{1}{2}AB = OA + \frac{1}{2}(OB - OA) = \frac{1}{2}(OA + OB)$

6. *Two forces of 10 N and 10 N act at 90°. Find the resultant.*

   ANSWER $10\sqrt{2} \approx 14.1\ N$ at $45°$ to each

7. *Explain why* $|a + b| \le |a| + |b|$ *always.*

   ANSWER The sum closes a triangle, and in any triangle one side is never longer than the other two together; equality only when the vectors point the same way.

07 Homework

Homework — 6 problems

Geometry 8, Темы 36–37, pp. 81–84. Draw the triangle for every addition.

1. *Find* $(4; -2) + (-7; 5)$ *and* $(4; -2) - (-7; 5)$.
2. *Find* $|(5; 12) + (0; 1)|$.
3. *Simplify AB + BC + CA.*
4. $OA = (2; -1)$, $OB = (6; 7)$. *Find AB and* $|AB|$.

5. *A swimmer heads north at 3 km/h in a current flowing east at 4 km/h. Find the actual speed.*
6. *In parallelogram ABCD with AB = a and AD = b, express AC and BD.*

Multiplying a vector by a number; the coordinates of a vector

Stretching, shrinking and reversing — and the test for parallel that comes free with it.

LESSON 42–43 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 38–39 | STAGE 9 · 13.2

LESSON CLOCK

12' | 24' | 24' | 12' | 8'

Double a vector on squared paper	12 min
Multiplying by a number	24 min
Coordinates of a vector	24 min
The collinearity test	12 min
Homework	8 min

LEARNING OBJECTIVES

- Multiply a vector by a scalar, and describe the effect on length and direction.
- Write a vector in coordinates from the coordinates of its ends.
- Test two vectors for collinearity.
- Use vectors to prove that a line is parallel to another.

TEXTBOOK

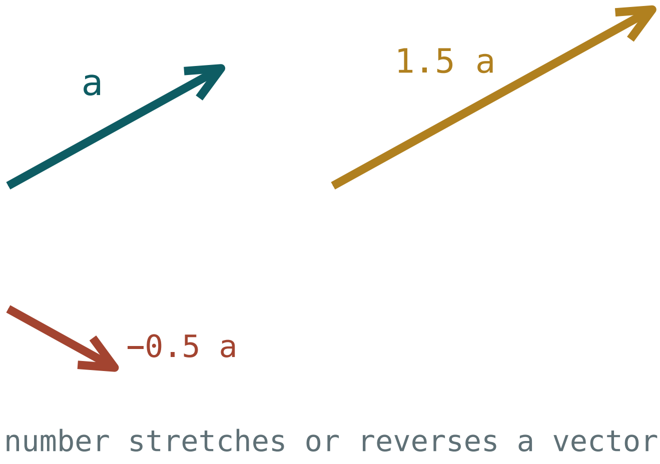
National	Темы 38–39, pp. 85–89
Cambridge	Learner's Book pp. 291–296
Workbook	Workbook 13.2

01 Explanation

Multiplying by a number

SCALAR MULTIPLE

$$k \cdot (a_1; a_2) = (ka_1; ka_2) \quad \text{and} \quad |k\boldsymbol{a}| = |k| \cdot |\boldsymbol{a}|$$



Multiplying by a number stretches or shrinks the arrow;
a negative number also turns it round.

K	EFFECT
$k > 1$	longer, same direction
$0 < k < 1$	shorter, same direction
$k = 0$	the zero vector
$k < 0$	direction reversed, length $\mid k \mid$ times

Note the modulus signs: $\mid -3\boldsymbol{a} \mid = 3 \mid \boldsymbol{a} \mid$. A length is never negative, however negative the multiplier.

Coordinates, and the test for parallel

The coordinates of a vector are found once and used everywhere:

$$AB = (x\_B - x\_A;\ \ y\_B - y\_A)$$

“End minus start” — and if you swap them you get the opposite vector, which is the commonest slip in this chapter.

TEST FOR COLLINEAR VECTORS

$\boldsymbol{a}$ and $\boldsymbol{b}$ are collinear exactly when $\boldsymbol{b} = k\boldsymbol{a}$ for some number k — that is, when

$$a_1b_2 - a_2b_1 = 0$$

Check $(2; 3)$ and $(6; 9)$: $2 \cdot 9 - 3 \cdot 6 = 0$ ✓, and indeed $(6; 9) = 3(2; 3)$. Check $(2; 3)$ and $(6; 8)$: $2 \cdot 8 - 3 \cdot 6 = -2 \neq 0$ — not collinear.

PARALLEL LINES FROM PARALLEL VECTORS

If AB and CD are collinear vectors and the four points are not all on one line, then the lines AB and CD are parallel. That is how vectors prove parallelism — no angles, no measuring.

02 Worked examples

EXAMPLE 1 **$\boldsymbol{a} = (-2; 5)$. Find $3\boldsymbol{a}$, $-\boldsymbol{a}$ and $|\,3\boldsymbol{a}\,|$.**

1. $3\boldsymbol{a} = (-6; 15)$

   Multiply each coordinate.

2. $-\boldsymbol{a} = (2; -5)$

   Reverse both signs.

3. $|\,\boldsymbol{a}\,| = \sqrt{4 + 25} = \sqrt{29}$

4. $|\,3\boldsymbol{a}\,| = 3\sqrt{29}$

   Three times as long.

ANSWER

$(-6; 15)$, $(2; -5)$, $3\sqrt{29}$

EXAMPLE 2 **$A(1; 2), B(4; 6), C(0; 1), D(6; 9)$. Is $AB \parallel CD$?**

1. $AB = (3; 4)$

   End minus start.

2. $CD = (6; 8)$

3. $3 \cdot 8 - 4 \cdot 6 = 0$

   The collinearity test.

4. $CD = 2 \cdot AB$

   So the vectors are parallel — and C is not on line AB.

ANSWER

Yes, $AB \parallel CD$ and CD is twice as long

EXAMPLE 3 **Find a unit vector in the direction of $(9; 12)$.**

1. $|(9; 12)| = \sqrt{81 + 144} = 15$

2. $\frac{1}{15} \cdot (9; 12)$

   Divide by the length.

3. $(0.6; 0.8)$

4. $\sqrt{0.36 + 0.64} = 1$

   Check: the length really is 1.

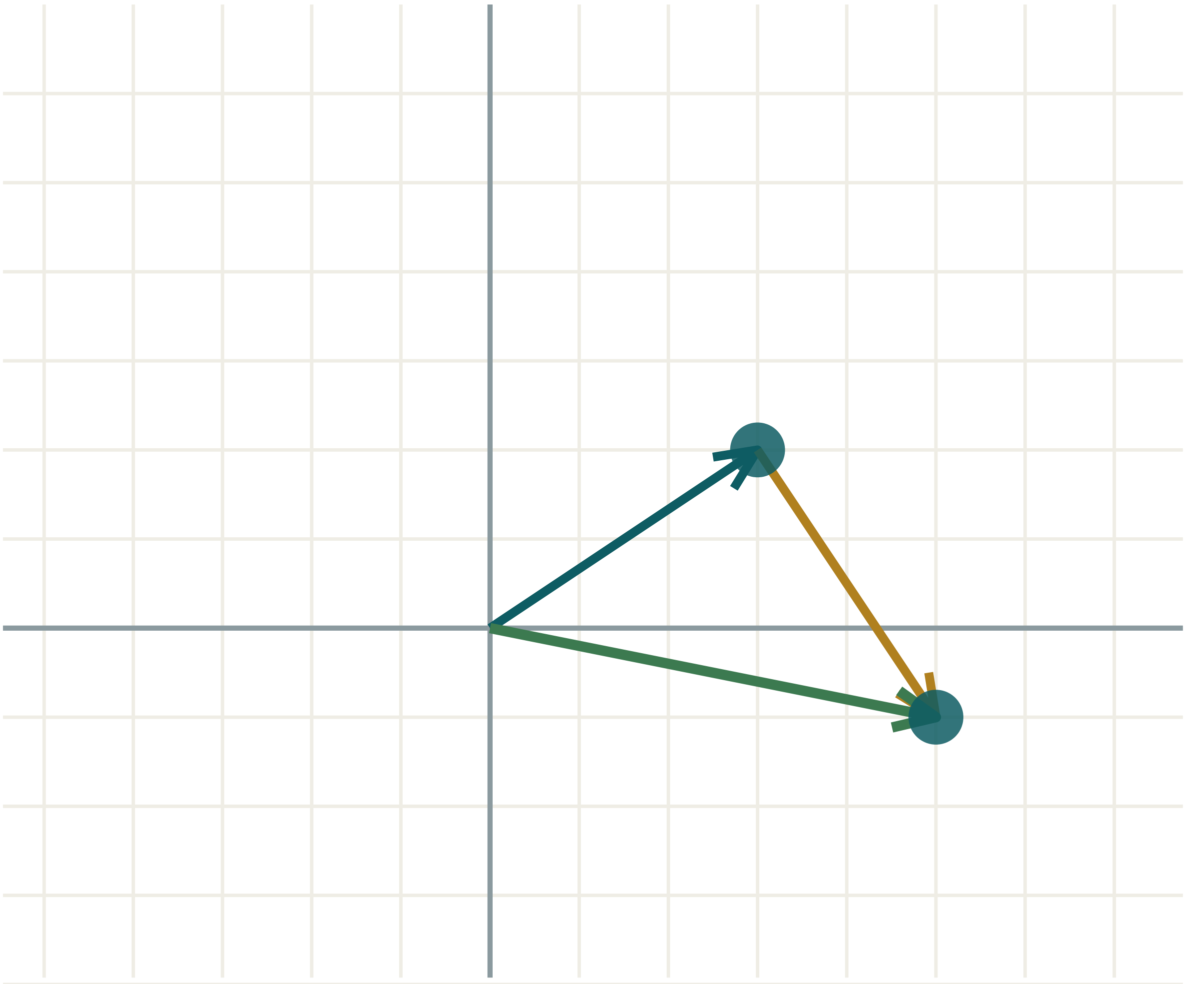
ANSWER

$(0.6; 0.8)$

03 Interactive model

Drag a tip and watch how each coordinate scales when the vector is stretched.

Scaling and adding vectors



a (3; 2) | b (2; -3) | a + b (5; -1) | | a | 3.61 | | b | 3.61 | | a + b | 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Scalar multiple	Skalyarga ko‘paytma	Умножение на число
Stretch	Cho‘zish	Растяжение
Compression	Siqish	Сжатие
Direction reversed	Yo‘nalish teskari	Направление противоположное
Coordinates of a vector	Vektor koordinatalari	Координаты вектора
Collinear	Kollinear	Коллинеарные
Unit vector	Birlik vektor	Единичный вектор
Position vector	Radius-vektor	Радиус-вектор
Parallel	Parallel	Параллельный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$4 \cdot (2; -3)$ is:

- $(8; -12)$
- $(6; 1)$
- $(8; 12)$
- $(2; -12)$

If $|\,\boldsymbol{a}\,| = 6$ then $|\,-2\boldsymbol{a}\,|$ is:

- -12
- 12
- 4
- 3

$(3; 5)$ and $(9; 15)$ are:

- perpendicular
- collinear
- equal
- opposite

For $A(2; 3)$ and $B(5; 1)$, AB is:

- $(3; -2)$
- $(-3; 2)$
- $(7; 4)$
- $(3; 2)$

A unit vector has length:

- 0
- 1
- *any*
- 10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find* $2 \cdot (3; 4)$.

   ANSWER $(6; 8)$

2. *Find* $-1 \cdot (5; -2)$.

   ANSWER $(-5; 2)$

3. *Find* $0.5 \cdot (8; 6)$.

   ANSWER $(4; 3)$

4. *If* $|a| = 4$, *find* $|3a|$.

   ANSWER 12

5. *Find* AB *for* $A(0; 0)$, $B(-3; 7)$.

   ANSWER $(-3; 7)$

6. *Are* $(1; 2)$ *and* $(2; 4)$ *collinear?*

   ANSWER Yes.

7. *Find* $|2 \cdot (3; 4)|$.

   ANSWER 10

Medium · the standard question

7 PROBLEMS

1. *Find* $3 \cdot (-2; 5) - 2 \cdot (1; 4)$.

   ANSWER $(-8; 7)$

2. *If* $|a| = 5$, *find* $|-4a|$.

   ANSWER 20

3. *Are* $(4; -6)$ *and* $(-6; 9)$ *collinear?*

   ANSWER Yes — $4 \cdot 9 - (-6)(-6) = 0$

4. *Are* $(2; 5)$ *and* $(4; 9)$ *collinear?*

   ANSWER No — $2 \cdot 9 - 5 \cdot 4 = -2 \neq 0$

5. $A(1; 1)$, $B(3; 5)$, $C(2; 0)$, $D(6; 8)$. *Is* $AB \parallel CD$?

   ANSWER Yes — $CD = 2 \cdot AB$

6. *Find a unit vector in the direction of* $(3; 4)$.

   ANSWER $(0.6; 0.8)$

7. *Find* k *so that* $(6; k)$ *is collinear with* $(2; 5)$.

   ANSWER $k = 15$

Hard · stretch and reason

7 PROBLEMS

1. *$a = (2; -1)$, $b = (-3; 4)$. Find $2a - 3b$ and its length.*

   ANSWER $(13; -14)$, length $\sqrt{365}$

2. *Find all k for which $(k; 6)$ and $(3; k)$ are collinear.*

   ANSWER $k^2 = 18$, $k = \pm 3\sqrt{2}$

3. *Find a vector of length 20 in the direction of $(-3; 4)$.*

   ANSWER $(-12; 16)$

4. *$A(0;0)$, $B(6;4)$. Find the point P on AB with $AP = \frac{2}{3} \cdot AB$.*

   ANSWER $P(4; \frac{8}{3})$

5. *M and N are midpoints of AB and AC. Show that $MN \parallel BC$ and $MN = \frac{1}{2}BC$, using vectors.*

   ANSWER $MN = AN - AM = \frac{1}{2}AC - \frac{1}{2}AB = \frac{1}{2}BC$ — a multiple of BC, so parallel and half as long

6. *$a = (1; 2)$ and $b = (3; k)$. Find k so that a and b are collinear, and then find b in terms of a.*

   ANSWER $k = 6$, $\boldsymbol{b} = 3\boldsymbol{a}$

7. *Explain why the collinearity test $a_1b_2 - a_2b_1 = 0$ works.*

   ANSWER If $\boldsymbol{b} = k\boldsymbol{a}$ then $a_1(ka_2) - a_2(ka_1) = 0$; conversely, when the expression is zero the ratios $\frac{b_1}{a_1}$ and $\frac{b_2}{a_2}$ agree, giving a common multiplier k.

07 Homework

Homework — 6 problems

Geometry 8, Темы 38–39, pp. 85–89. Show the collinearity test in full.

1. *Find $5 \cdot (-1; 3)$ and its length.*
2. *If $|a| = 7$, find $|-3a|$.*

3. *Are $(6; -4)$ and $(-9; 6)$ collinear? Justify.*
4. *Find k so that $(4; k)$ is collinear with $(3; 12)$.*
5. *Find a unit vector in the direction of $(5; 12)$.*
6. *$A(1;0)$, $B(4;6)$, $C(0;2)$, $D(2;6)$. Is $AB \parallel CD$?*

Operations with vectors in coordinates; applications

Everything from the last four lessons, applied to real proofs and real journeys.

LESSON 44–45 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 40–41 | STAGE 9 · 13.1–13.2

LESSON CLOCK

12' | 24' | 24' | 12' | 8'

Recall the four operations	12 min
Position vectors	24 min
Vector proofs	24 min
Journeys and velocities	12 min
Homework	8 min

LEARNING OBJECTIVES

- Combine addition, subtraction and scalar multiplication in coordinates.
- Use position vectors to find midpoints and points dividing a segment.
- Prove geometric statements with vectors.
- Solve displacement and velocity problems.

TEXTBOOK

National	Темы 40–41, pp. 90–95
Cambridge	Learner’s Book pp. 284–296
Workbook	Workbook 13.2

01 Explanation

Position vectors

The **position vector** of A is OA, drawn from the origin. If $A(x; y)$ then $OA = (x; y)$ — a point and its position vector carry the same two numbers. From that one identification everything follows:

$$AB = OB - OA \;\mid\; OM = \frac{1}{2}(OA + OB) \text{ for the midpoint } M \text{ of } AB$$

To reach a point P that divides AB in a given proportion, start at A and travel part of the way:

$$OP = OA + t \cdot AB, \text{ where } t = 0 \text{ is } A \text{ and } t = 1 \text{ is } B$$

$t = \frac{1}{2}$ gives the midpoint, $t = \frac{1}{3}$ gives the point one third of the way, and so on.

Proving with vectors

A vector proof has three moves, and only three:

1. Name two vectors as $\boldsymbol{a}$ and $\boldsymbol{b}$ — usually two sides from one vertex.
2. Write every other vector in the figure in terms of them.
3. Read the conclusion: a multiple means **parallel**; equal vectors mean **equal and parallel**; $\boldsymbol{0}$ means the path closes.

THE MIDLINE THEOREM, IN THREE LINES

Let $AB = \boldsymbol{c}$ and $AC = \boldsymbol{b}$, with M and N the midpoints of AB and AC.

$$MN = AN - AM = \frac{1}{2}\boldsymbol{b} - \frac{1}{2}\boldsymbol{c} = \frac{1}{2}(\boldsymbol{b} - \boldsymbol{c}) = \frac{1}{2}BC$$

A multiple of BC, so $MN \parallel BC$; the multiplier is $\frac{1}{2}$, so $MN = \frac{1}{2}BC$. The whole of Quarter I's midline theorem, in one line of algebra.

Journeys and velocities

Displacements add as vectors, so a journey with several stages is one sum:

$$\text{walk } (3;\, 0), \text{ then } (0;\, 4), \text{ then } (-1;\, 2) \Longrightarrow \text{total } (2;\, 6)$$

The total **displacement** is $(2; 6)$, of length $\sqrt{40} \approx 6.3$, while the **distance walked** is $3 + 4 + \sqrt{5} \approx 9.2$. They are different quantities and questions ask for both.

Velocities add in the same way. A boat's velocity through the water plus the current's velocity gives the velocity over the ground — which is why a boat aiming straight across a river never arrives straight across.

02 Worked examples

EXAMPLE 1

$OA = (1; 4)$, $OB = (7; 2)$. Find AB, the midpoint of AB, and the point one third of the way from A to B.

1. $AB = OB - OA = (6; -2)$

   End minus start.

2. $OM = \frac{1}{2}((1; 4) + (7; 2)) = (4; 3)$

   Average the position vectors.

3. $OP = OA + \frac{1}{3}AB$

4. $= (1; 4) + (2; -\frac{2}{3}) = (3; 3\frac{1}{3})$

ANSWER

$AB = (6; -2)$, $M(4; 3)$, $P(3; 3\frac{1}{3})$

EXAMPLE 2 **A cyclist rides 6 km east, then 8 km north. Find the total distance and the displacement.**

1. $(6; 0) + (0; 8) = (6; 8)$

   Add the two displacements.

2. $|\,(6; 8)\,| = 10\ km$

   The displacement.

3. $6 + 8 = 14\ km$

   The distance actually ridden.

4. The two differ because the route bends.

ANSWER

Distance $14\ km$, displacement $10\ km$ north-east

EXAMPLE 3 **Use vectors to prove that the diagonals of a parallelogram bisect each other.**

1. $AB = \boldsymbol{a}, AD = \boldsymbol{b}$

   Name two sides from one vertex.

2. $AC = \boldsymbol{a} + \boldsymbol{b}$

   The diagonal from A.

3. Midpoint of AC: $\frac{1}{2}(\boldsymbol{a} + \boldsymbol{b})$ from A.

4. Midpoint of BD: $\boldsymbol{a} + \frac{1}{2}(\boldsymbol{b} - \boldsymbol{a}) = \frac{1}{2}(\boldsymbol{a} + \boldsymbol{b})$.

   The same point.

5. The diagonals cross at their common midpoint.

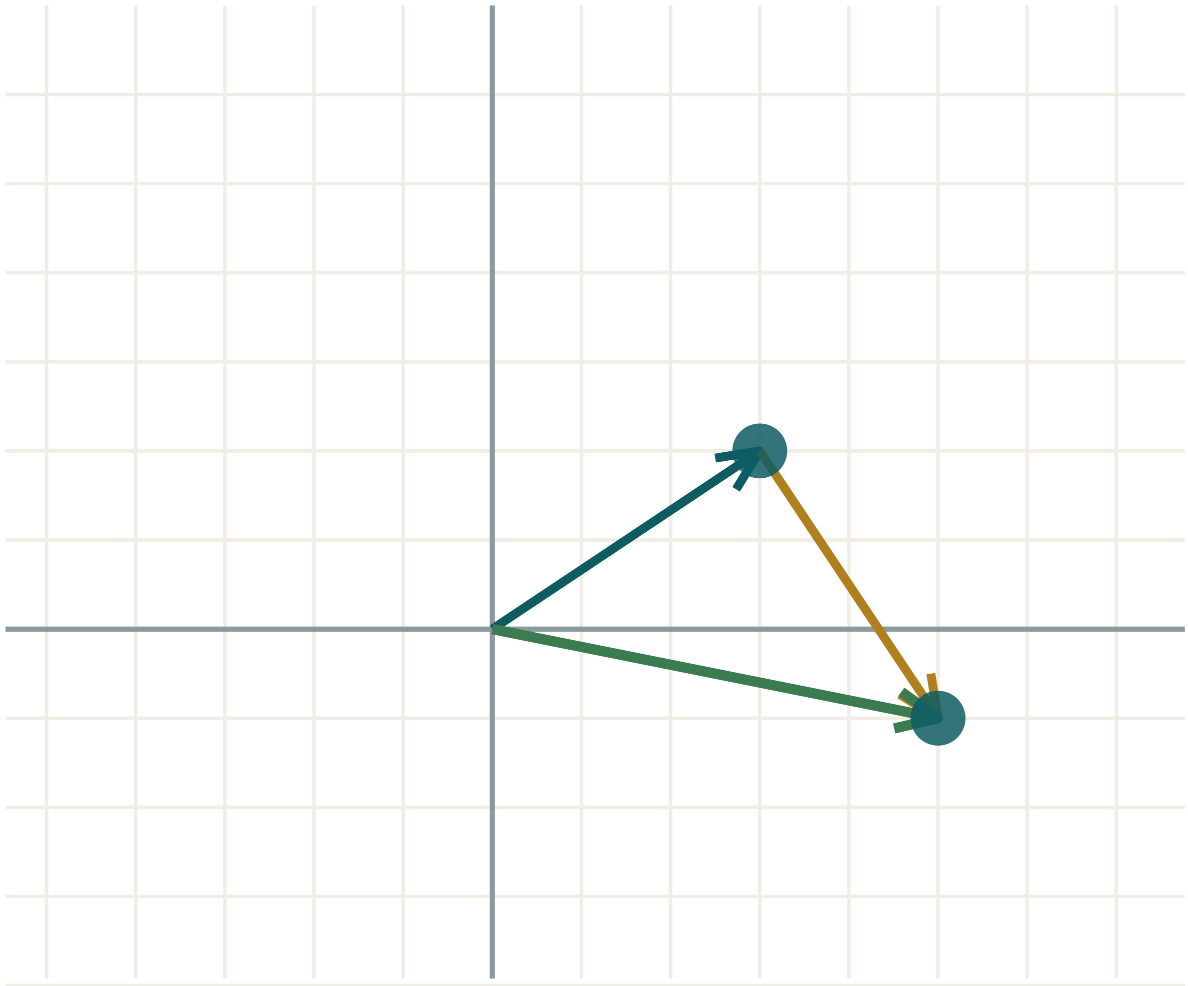
ANSWER

Proved

03 Interactive model

Build a journey out of two vectors and read the resultant off the green arrow.

Two stages of a journey



a (3; 2)	b (2; -3)	a + b (5; -1)	|a| 3.61	|b| 3.61	|a + b| 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Position vector	Radius-vektor	Радиус-вектор
Resultant displacement	Natijaviy siljish	Результирующее перемещение
Velocity	Tezlik (vektor)	Скорость (вектор)
Component	Tashkil etuvchi	Составляющая
Midpoint (vector form)	O‘rta nuqta (vektor shakli)	Середина (в векторах)
Divides in the ratio	Nisbatda bo‘ladi	Делит в отношении
Vector proof	Vektorli isbot	Векторное доказательство
Centroid	Medianalar kesishish nuqtasi	Центроид (точка пересечения медиан)
Bearing	Azimut	Азимут

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

If $OA = (2; 1)$ and $OB = (8; 5)$ then AB is:

- $(10; 6)$
- $(6; 4)$
- $(-6; -4)$
- $(4; 6)$

The midpoint of $A(1; 3)$ and $B(7; 9)$ has position vector:

- $(8; 12)$
- $(4; 6)$
- $(6; 6)$
- $(3; 4)$

Walk $(4; 0)$ then $(0; 3)$. The displacement is:

- 7
- 5
- 12
- 1

If $MN = \frac{1}{2} BC$ then MN is:

- perpendicular to BC
- parallel to BC and half its length
- equal to BC
- unrelated

The distance walked and the displacement are equal only when:

- always
- the route is a straight line in one direction
- the route is closed
- never

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $OA = (1; 1)$, $OB = (4; 5)$. Find AB.

   ANSWER $(3; 4)$

2. Find the midpoint of $A(0; 0)$ and $B(6; 10)$.

   ANSWER $(3; 5)$

3. Walk $(3; 0)$ then $(0; 4)$. Find the displacement.

   ANSWER 5

4. Find $2 \cdot (1; 3) + (4; -2)$.

   ANSWER $(6; 4)$

5. $OA = (5; 2)$. Write the coordinates of A.

   ANSWER $A(5; 2)$

6. Walk $(2; 0)$ then $(-2; 0)$. Find the displacement.

   ANSWER 0

7. Find $|(9; 12)|$.

   ANSWER 15

Medium · the standard question

7 PROBLEMS

1. $OA = (-1; 3)$, $OB = (5; -1)$. Find AB and its length.

   ANSWER $(6; -4)$, $\sqrt{52} = 2\sqrt{13}$

2. Find the midpoint of $A(-3; 4)$ and $B(9; -2)$.

   ANSWER $(3; 1)$

3. A cyclist rides 5 km north then 12 km east. Find distance and displacement.

   ANSWER 17 km and 13 km

4. Find $3 \cdot (2; -1) - 2 \cdot (4; 1)$.

   ANSWER $(-2; -5)$

5. $OA = (2; 2)$, $AB = (6; 4)$. Find OB.

   ANSWER $(8; 6)$

6. Find the point one quarter of the way from $A(0; 0)$ to $B(8; 12)$.

   ANSWER $(2; 3)$

7. A plane flies at $(0; 300)$ in a wind of $(40; 0)$. Find its ground speed.

   ANSWER $\sqrt{91600} \approx 303$ km/h

Hard · stretch and reason

7 PROBLEMS

1. *$OA = (1; 2)$, $OB = (7; 5)$, $OC = (4; 11)$. Find the centroid of triangle ABC.*

   ANSWER $(4; 6)$ — the average of the three position vectors

2. *$A(1; 1)$, $B(9; 5)$. Find P on AB with $AP : PB = 3 : 1$.*

   ANSWER $P(7; 4)$

3. *Prove with vectors that the midpoints of the sides of a quadrilateral form a parallelogram.*

   ANSWER With position vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}$, the midpoint figure has opposite sides $\frac{1}{2}(\boldsymbol{c} - \boldsymbol{a})$ twice over — equal vectors, so a parallelogram.

4. *A boat can do 5 km/h. It heads north across a river flowing east at 3 km/h. Find its speed and how far downstream it lands after crossing 400 m.*

   ANSWER $\sqrt{34} \approx 5.83$ km/h; it drifts 240 m downstream

5. *Walk $(4; 3)$, then $(-1; 5)$, then $(-3; -8)$. Find the total displacement.*

   ANSWER $(0; 0)$ — the walk returns to its start

6. *$OA = (2; 3)$ and $M(5; 7)$ is the midpoint of AB. Find OB.*

   ANSWER $(8; 11)$

7. *Prove with vectors that a quadrilateral with one pair of opposite sides equal and parallel is a parallelogram.*

   ANSWER If $AB = DC$ as vectors then $AD = AB + BD = DC + BD = BC$, so the other pair is equal and parallel too.

07 Homework

Homework — 6 problems

Geometry 8, Темы 40–41, pp. 90–95. Name your two base vectors before starting any proof.

1. *$OA = (3; -2)$, $OB = (9; 6)$. Find AB, $|AB|$ and the midpoint of AB.*
2. *Find $4 \cdot (1; -2) - 3 \cdot (2; 1)$.*

3. *A walker goes 8 km east then 6 km north. Find distance and displacement.*
4. *$A(2; 1)$, $B(10; 9)$. Find P on AB with $AP : PB = 1 : 3$.*
5. *$OA = (1; 5)$ and $M(4; 2)$ is the midpoint of AB. Find OB.*
6. *Use vectors to show that the midline of a triangle is parallel to the third side.*

The concept of area; the area of a rectangle

Four properties that every area must have — and the one formula everything else is built on.

LESSON 46 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 45–46 | STAGE 9 · 7.1

LESSON CLOCK

5' | 12' | 12' | 7' | 4'

Count squares to find an area	5 min
The four properties	12 min
Rectangle and square	12 min
Units and conversions	7 min
Homework	4 min

LEARNING OBJECTIVES

- State the four properties of area.
- Use the formula for the area of a rectangle and of a square.
- Convert between units of area.
- Find a missing side from a given area.

TEXTBOOK

National	Темы 45–46, pp. 100–105
Cambridge	Learner's Book pp. 142–148
Workbook	Workbook 7.1

01 Explanation

What area is

Area measures how much of the plane a figure covers, in **unit squares**. Whatever definition is used, it must obey four rules:

PROPERTIES OF AREA

1. Area is a **positive** number.
2. **Congruent** figures have **equal** areas.
3. If a figure is cut into parts, the areas of the parts **add up** to the whole.
4. The area of a square of side 1 is 1.

Property 3 — **additivity** — is the workhorse. It is why a trapezium can be split into a rectangle and two triangles, and why an L-shaped room can be measured as two rectangles.

EQUAL AREAS ≠ CONGRUENT

A 2×8 rectangle and a 4×4 square both have area 16, but you cannot lay one on the other. Rule 2 works in one direction only.

Rectangle, square, and units

$$S_{rectangle} = a \cdot b \mid S_{square} = a^2$$

The formula is really just counting: a 5×3 rectangle holds three rows of five unit squares.

UNIT	IN M²	USED FOR
$1\ cm^2$	0.0001	a page, a tile
$1\ m^2$	1	a room, a carpet
$1\ a$ (are)	100	a garden plot
$1\ ha$	$10\ 000$	a field
$1\ km^2$	$1\ 000\ 000$	a district

Lengths scale by k; areas scale by k^2. $1\ m = 100\ cm$, so $1\ m^2 = 10\ 000\ cm^2$ — a fact that catches out more pupils than any other in this chapter.

02 Worked examples

EXAMPLE 1

A rectangular room is $4.5\ m$ by $3.2\ m$. Find its area, and the cost of carpet at $85\ 000$ so‘m per m².

1. $S = 4.5 \cdot 3.2 = 14.4\ m^2$
2. $14.4 \cdot 85\ 000$
3. $= 1\ 224\ 000$ so‘m

ANSWER

$14.4\ m^2$, $1\ 224\ 000$ so‘m

EXAMPLE 2 **A rectangle has area $84\ cm^2$ and one side $7\ cm$. Find its perimeter.**

1. $84 \div 7 = 12$

   The other side.

2. $P = 2(7 + 12)$

3. $= 38\ cm$

ANSWER

$38\ cm$

EXAMPLE 3 **An L-shaped floor is a 6×4 rectangle with a 2×2 square cut from one corner. Find its area.**

1. $6 \cdot 4 = 24$

   The whole rectangle.

2. $2 \cdot 2 = 4$

   The corner removed.

3. $24 - 4 = 20$

   Additivity, used backwards.

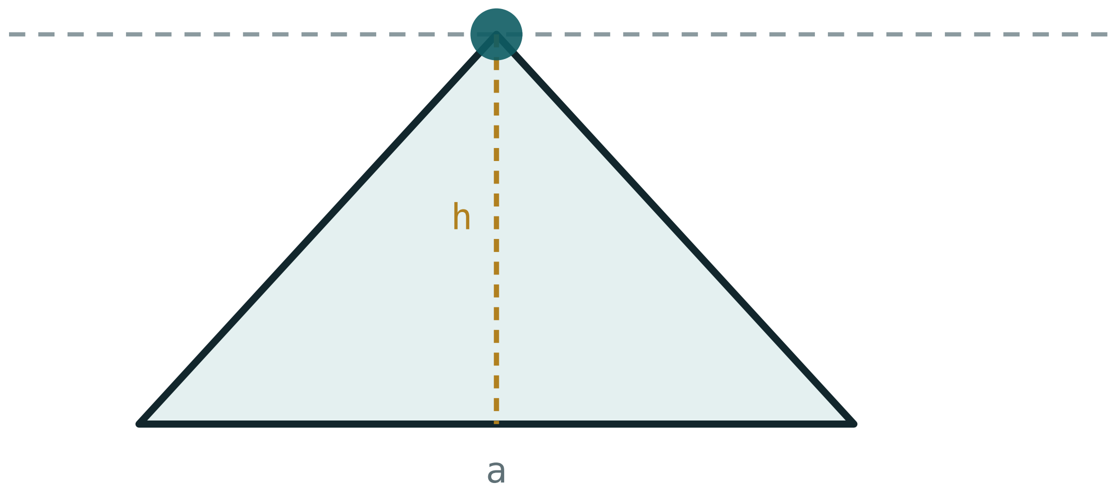
ANSWER

$20\ m^2$

03 Interactive model

Change the base and height and watch the area follow the product.

Base times height



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Area	Yuza	Площадь
Unit square	Birlik kvadrat	Единичный квадрат
Congruent figures	Teng figuralar	Равные фигуры
Additivity of area	Yuzaning additivligi	Аддитивность площади
Rectangle	To‘g‘ri to‘rtburchak	Прямоугольник
Square	Kvadrat	Квадрат
Square metre	Kvadrat metr	Квадратный метр
Hectare	Gektar	Гектар
Dimensions	O‘lchamlar	Размеры

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The area of a 7×9 rectangle is:

- 16
- 32
- 63
- 72

$1\ m^2$ equals:

- $100\ cm^2$
- $1\ 000\ cm^2$
- $10\ 000\ cm^2$
- $1\ 000\ 000\ cm^2$

A square of area $49\ cm^2$ has perimeter:

- $14\ cm$
- $28\ cm$
- $49\ cm$
- $196\ cm$

Congruent figures:

- may have different areas
- always have equal areas
- always are rectangles
- have equal perimeters but not areas

1 ha equals:

- $100\ m^2$
- $1\ 000\ m^2$
- $10\ 000\ m^2$
- $100\ 000\ m^2$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the area of a 5 cm by 8 cm rectangle.*

   ANSWER *40 cm²*

2. *Find the area of a square of side 9 cm.*

   ANSWER *81 cm²*

3. *Convert 3 m² to cm².*

   ANSWER *30 000 cm²*

4. *A rectangle has area 36 and one side 4. Find the other.*

   ANSWER *9*

5. *Find the area of a 2.5 m by 4 m rectangle.*

   ANSWER *10 m²*

6. *Convert 2 ha to m².*

   ANSWER *20 000 m²*

7. *Find the side of a square of area 144 cm².*

   ANSWER *12 cm*

Medium · the standard question

7 PROBLEMS

1. *A room 5.5 m by 4 m is carpeted at 90 000 so‘m/m². Find the cost.*

   ANSWER *1 980 000 so‘m*

2. *A rectangle has area 150 cm^2 and perimeter 50 cm. Find its sides.*

   ANSWER *10 cm* and *15 cm*

3. *An L-shape is a 8×5 rectangle less a 3×2 corner. Find its area.*

   ANSWER *34*

4. *Convert 45 000 cm^2 to m^2.*

   ANSWER *4.5 m^2*

5. *A square field has area 1 ha. Find its side.*

   ANSWER *100 m*

6. *A rectangle 12 cm by 5 cm is cut into two equal squares plus a rectangle. Is that possible? Explain.*

   ANSWER Two 5×5 squares use *10 cm* of the length, leaving a 2×5 rectangle — yes.

7. *The sides of a rectangle are doubled. What happens to the area?*

   ANSWER It becomes four times as large.

Hard · stretch and reason

7 PROBLEMS

1. *A rectangle has perimeter 40 cm. Find the dimensions giving the greatest area.*

   ANSWER 10×10 — the square, area $100\ cm^2$

2. *A path 1 m wide runs round the outside of a 10 m by 6 m garden. Find the area of the path.*

   ANSWER $12 \cdot 8 - 10 \cdot 6 = 36\ m^2$

3. *A rectangle has area 60 cm² and its length is 4 cm more than its width. Find its sides.*

   ANSWER $6\ cm$ by $10\ cm$

4. *A field of 2.4 ha is rectangular with one side 120 m. Find the other side and the perimeter.*

   ANSWER $200\ m$, perimeter $640\ m$

5. *A floor 4.2 m by 3.6 m is tiled with 30 cm squares. How many tiles are needed?*

   ANSWER $14 \cdot 12 = 168$

6. *The area of a square is numerically equal to its perimeter. Find its side.*

   ANSWER $a^2 = 4a$, so $a = 4$

7. *Explain why two figures with equal areas need not be congruent.*

   ANSWER Area is a single number and loses all information about shape; a 1×12 rectangle and a 3×4 rectangle share an area but no dimension.

07 Homework

Homework — 5 problems

Geometry 8, Темы 45–46, pp. 100–105. Always write the unit.

1. *Find the area of a 7.5 m by 6 m room.*
2. *Convert 5 m² to cm² and 30 000 cm² to m².*
3. *A rectangle has area 96 cm² and one side 8 cm. Find its perimeter.*
4. *An L-shape is a 10×7 rectangle less a 4×3 corner. Find its area.*

5. *A square has area 225 m^2. Find its side and perimeter.*

Home / Grade 8 / Geometry / Quarter 3

The area of a parallelogram and of a triangle

Cut a corner off and slide it across — and the parallelogram becomes a rectangle.

LESSON 47–48 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 47–48 | STAGE 9 · 7.1–7.2

LESSON CLOCK

12' | 28' | 24' | 8' | 8'

Cut and slide a paper parallelogram	12 min
The parallelogram formula	28 min
The triangle formula	24 min
Two sides and an angle	8 min
Homework	8 min

LEARNING OBJECTIVES

- Derive and use $S = a \cdot h$ for a parallelogram.
- Derive and use $S = ½\, a \cdot h$ for a triangle.
- Use $S = ½\, ab \sin C$ when two sides and the included angle are known.
- Recognise that triangles on the same base between the same parallels have equal areas.

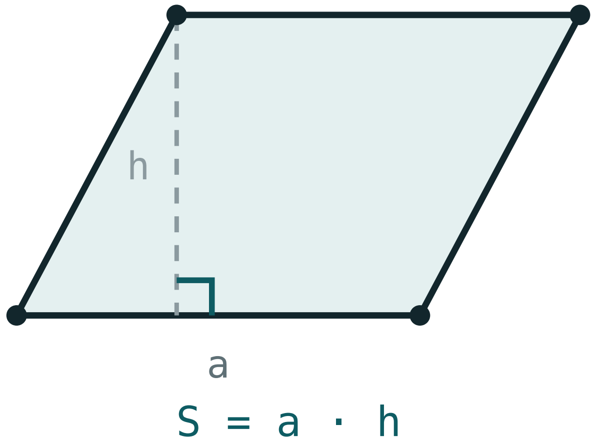
TEXTBOOK

National	Темы 47–48, pp. 106–113
Cambridge	Learner's Book pp. 142–155
Workbook	Workbook 7.1–7.2

01 Explanation

The parallelogram

Drop a perpendicular from one vertex, cut off the right triangle it makes and slide it to the other end. Nothing is lost and nothing is added — the parallelogram has become a rectangle with the same base and the same height.



Cut the triangle off one end and slide it to the other: the parallelogram becomes a rectangle of the same base and height.

AREA OF A PARALLELOGRAM

$$S = a \cdot h_a$$

where h_a is the height **to the side *a***.

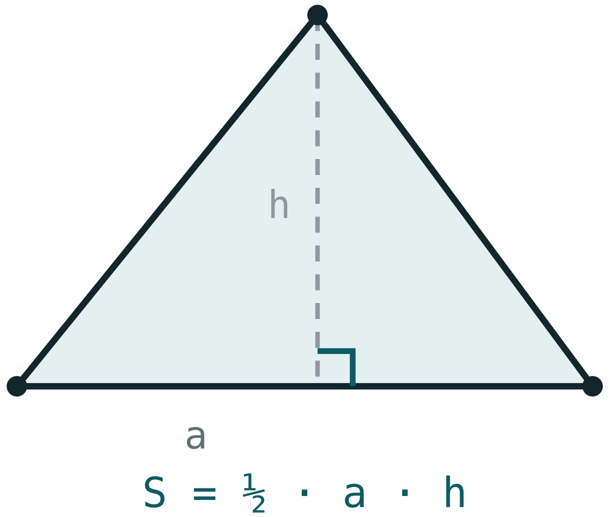
HEIGHT, NOT SLANT SIDE

In a parallelogram with sides 8 and 5 the area is **not** 40 unless the angle is right. The height is the perpendicular distance between the two parallel sides, and it is always shorter than the slant side.

Since $S = a \cdot h_a = b \cdot h_b$, a longer side always has a shorter height. That single equation solves a whole family of problems.

The triangle

Any triangle is exactly half of a parallelogram: copy it, turn the copy through $180°$ and join the two along a side.



Two copies of a triangle make a parallelogram, so a triangle is half of base times height.

$$S = \frac{1}{2}\ a \cdot h_a$$

For a right-angled triangle the two legs serve as base and height, so $S = \frac{1}{2}ab$ immediately.

TWO SIDES AND THE ANGLE BETWEEN THEM

$$S = \frac{1}{2}\ ab \cdot \sin C$$

because the height to a is $b \sin C$. With $C = 90°$ this is the right-angle formula again, since $\sin 90° = 1$.

Same base, same parallels

The height of a triangle does not depend on where the apex sits along a line parallel to the base. So:

EQUAL AREAS

Triangles with the **same base** and apexes on the **same line parallel** to it have **equal areas** — however different they look.

Two consequences worth remembering:

- A **median** divides a triangle into two of equal area — the two halves share a height and have equal bases.
- The diagonals of a parallelogram cut it into four triangles of equal area.

02 Worked examples

EXAMPLE 1 **A parallelogram has sides *10 cm* and *6 cm*, and the height to the *10 cm* side is *4 cm*. Find its area and the other height.**

1. $S = 10 \cdot 4 = 40\ cm^2$

   Base times its own height.

2. $40 = 6 \cdot h_b$

   The same area, measured from the other side.

3. $h_b = \dfrac{40}{6} \approx 6.67\ cm$

   Shorter side, taller height.

ANSWER

$40\ cm^2,\ h_b \approx 6.67\ cm$

EXAMPLE 2 **Find the area of a triangle with sides *7 cm* and *8 cm* and an included angle of *30°*.**

1. $S = \frac{1}{2} \cdot 7 \cdot 8 \cdot \sin 30°$
2. $\sin 30° = 0.5$
3. $S = \frac{1}{2} \cdot 56 \cdot 0.5$
4. $= 14\ cm^2$

ANSWER

$14\ cm^2$

EXAMPLE 3 **A triangle has area *54 cm²* and base *12 cm*. Find its height, and the area of each half made by the median to that base.**

1. $54 = \frac{1}{2} \cdot 12 \cdot h$
2. $54 = 6h,\ h = 9\ cm$
3. $54 \div 2 = 27\ cm^2$

   A median always halves the area.

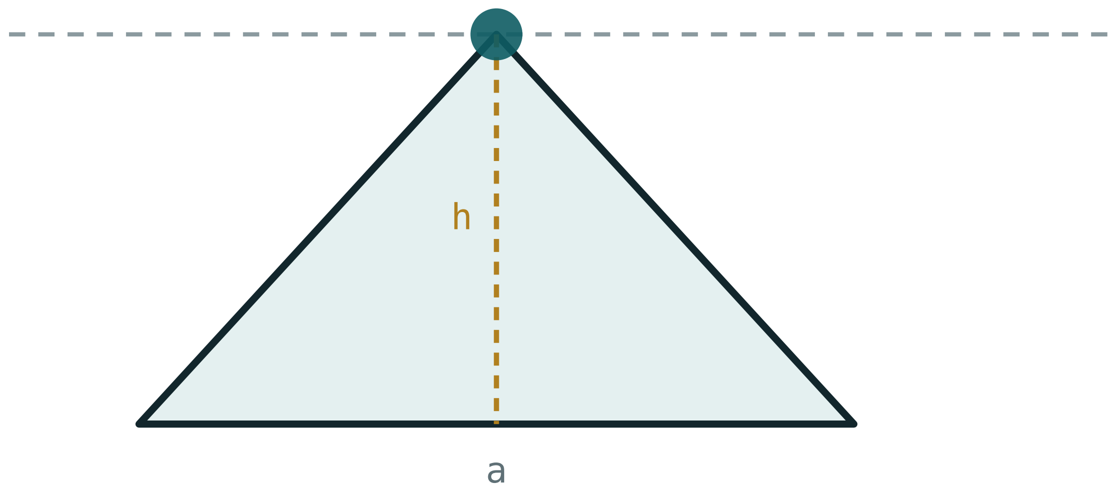
ANSWER

$h = 9\ cm$; each half $27\ cm^2$

03 Interactive model

Slide the apex along the top line: the shape changes completely, the area does not.

Same base, same height, same area



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Parallelogram	Parallelogramm	Параллелограмм
Base	Asos	Основание
Height (altitude)	Balandlik	Высота
Corresponding height	Mos balandlik	Соответствующая высота
Triangle	Uchburchak	Треугольник
Included angle	Orasidagi burchak	Угол между сторонами
Equal areas	Teng yuzalar	Равновеликие
Median	Mediana	Медиана
Between the same parallels	Bir xil parallellar orasida	Между теми же параллельными

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A parallelogram with base 9 and height 4 has area:

- 13
- 18
- 36
- 72

A triangle with base 10 and height 6 has area:

- 60
- 30
- 16
- 15

A parallelogram has sides 8 and 5. Its area is:

- 40
- 26
- less than 40
- 20

A median divides a triangle into two parts of:

- unequal area
- equal area
- equal perimeter
- equal angles

For sides 6 and 10 with an included angle of $30°$, the area is:

- 30
- 15
- 60
- 7.5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the area of a parallelogram with base 7 and height 5.*

   ANSWER *35*

2. *Find the area of a triangle with base 8 and height 3.*

   ANSWER *12*

3. *Find the area of a right triangle with legs 6 and 9.*

   ANSWER *27*

4. *A triangle has area 20 and base 10. Find its height.*

   ANSWER *4*

5. *A parallelogram has area 48 and base 12. Find its height.*

   ANSWER *4*

6. *Find the area of a triangle with base 5 and height 5.*

   ANSWER *12.5*

7. *A median divides a triangle of area 36. Find each part.*

   ANSWER *18*

Medium · the standard question

7 PROBLEMS

1. *A parallelogram has sides 12 and 9, and the height to the 12-side is 6. Find the other height.*

   ANSWER *8*

2. *Find the area of a triangle with sides 10 and 12 and included angle 30°.*

   ANSWER *30*

3. *A triangle has area 42 cm² and height 7 cm. Find its base.*

   ANSWER *12 cm*

4. *The diagonals of a parallelogram of area 60 cut it into four triangles. Find each area.*

   ANSWER *15*

5. *Find the area of a triangle with vertices (0;0), (6;0), (0;8).*

   ANSWER *24*

6. *A parallelogram has base 15 cm and area 90 cm². Find its height.*

   ANSWER *6 cm*

7. *Find the area of a triangle with sides 8 and 8 and included angle 90°.*

   ANSWER *32*

Hard · stretch and reason

7 PROBLEMS

1. *A parallelogram has sides 10 and 8 with an angle of 30°. Find its area.*

   ANSWER $10 \cdot 8 \cdot \sin 30° = 40$

2. *A triangle has sides 13, 14, 15. Find its area.*

   ANSWER 84 — the height to the 14-side is 12

3. *The area of a triangle is 24 cm² and two of its sides are 8 cm and 12 cm. Find the angle between them.*

   ANSWER $\sin C = 0.5$, so $C = 30°$ (or $150°$)

4. *ABCD is a parallelogram and P is any point on AB. Show that the area of triangle PCD is half the area of ABCD.*

   ANSWER The triangle has base CD and its apex lies on the parallel line AB, so its height is the height of the parallelogram; its area is $\frac{1}{2} \cdot CD \cdot h$.

5. *Find the area of a triangle with vertices (1;1), (5;2), (3;7).*

   ANSWER 11 — box $4 \cdot 6 = 24$ less three corner triangles

6. *A parallelogram has diagonals of 10 and 6 crossing at 90°. Find its area.*

   ANSWER 30 — it is a rhombus, area $\frac{1}{2} d_1 d_2$

7. *Two triangles have the same base and equal areas. Must their apexes lie on a line parallel to that base?*

   ANSWER Yes — equal area with equal base forces equal height, so both apexes are the same distance from the base line, on one of the two parallels.

07 Homework

Homework — 6 problems

Geometry 8, Темы 47–48, pp. 106–113. Mark the height on every diagram.

1. *Find the area of a parallelogram with base 14 and height 9.*

2. *Find the area of a triangle with base 16 and height 7.5.*
3. *A parallelogram has sides 15 and 10; the height to the 15-side is 8. Find the other height.*
4. *Find the area of a triangle with sides 9 and 14 and included angle 30°.*
5. *A triangle has area 63 cm² and base 18 cm. Find its height.*
6. *Find the area of the triangle with vertices (0;0), (10;0), (4;6).*

The area of a rhombus and of a trapezium

Two formulas, both of them the parallelogram formula in disguise.

LESSON 49 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 49–50 | STAGE 9 · 7.2

LESSON CLOCK

5' 12' 14' 5' 4'

Recall the rhombus properties	5 min
Area of a rhombus	12 min
Area of a trapezium	14 min
Working backwards	5 min
Homework	4 min

LEARNING OBJECTIVES

- Use $S = ½\, d_1 d_2$ for a rhombus.
- Use $S = ½\,(a + b)\, h$ for a trapezium.
- Explain why each formula works.
- Find a missing length from a given area.

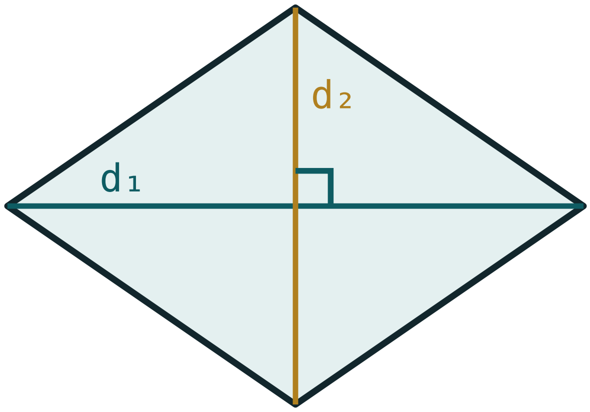
TEXTBOOK

National	Темы 49–50, pp. 114–118
Cambridge	Learner's Book pp. 149–155
Workbook	Workbook 7.2

01 Explanation

The rhombus

A rhombus is a parallelogram, so $S = a \cdot h$ still works. But its diagonals are perpendicular and bisect each other, and that gives a second formula which is usually more convenient:



$$S = \tfrac{1}{2} \cdot d_1 \cdot d_2$$

The perpendicular diagonals cut the rhombus into four congruent right triangles.

AREA OF A RHOMBUS

$$S = \frac{1}{2}\, d_1 \cdot d_2$$

Why: the four right triangles each have legs $\frac{d_1}{2}$ and $\frac{d_2}{2}$, so each has area $\frac{d_1 d_2}{8}$, and four of them make $\frac{d_1 d_2}{2}$.

The same formula holds for any quadrilateral whose diagonals are perpendicular — a kite, for instance — and for a square, where $d_1 = d_2$.

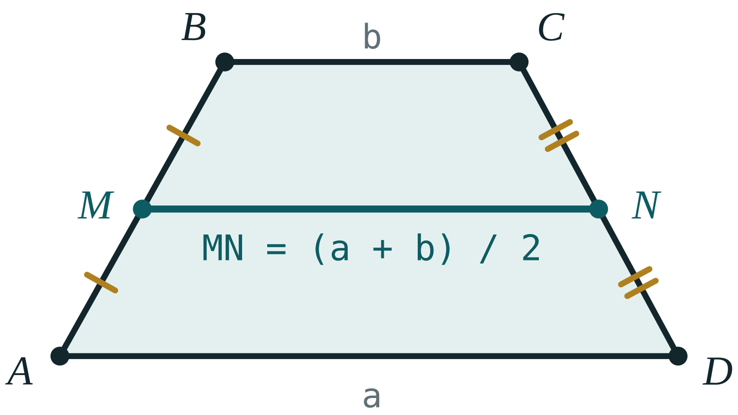
The trapezium

Copy the trapezium, turn the copy through $180°$ and join the two along a leg. The result is a parallelogram of base $a + b$ and height h. The trapezium is half of it:

AREA OF A TRAPEZIUM

$$S = \frac{a + b}{2} \cdot h$$

— the average of the two parallel sides, times the height.



The midline of a trapezium is the average of the two parallel sides, so the area is midline × height.

Since the midline $m = \frac{a+b}{2}$, the formula reads simply $S = m \cdot h$: area is midline times height, exactly as for a rectangle.

> AVERAGE, THEN MULTIPLY
>
> The commonest error is $(a + b) \cdot h$ without the halving — an answer exactly twice too big. Sanity-check against the enclosing rectangle: a trapezium must be smaller than $b \cdot h$ when b is the longer parallel side.

02 Worked examples

EXAMPLE 1 **A rhombus has diagonals *12 cm* and *16 cm*. Find its area, its side and its height.**

1. $S = \dfrac{1}{2} \cdot 12 \cdot 16 = 96\ cm^2$
2. *half-diagonals 6 and 8*
3. $a = \sqrt{36 + 64} = 10\ cm$

   Pythagoras in one of the four right triangles.
4. $96 = 10 \cdot h,\ h = 9.6\ cm$

   Using the parallelogram formula.

ANSWER

$96\ cm^2$, side $10\ cm$, height $9.6\ cm$

EXAMPLE 2 **A trapezium has parallel sides *8 cm* and *14 cm* and height *5 cm*. Find its area and its midline.**

1. $m = \dfrac{8 + 14}{2} = 11\ cm$

   The midline.
2. $S = 11 \cdot 5$
3. $= 55\ cm^2$

   Less than $14 \cdot 5 = 70$ ✓

ANSWER

$55\ cm^2$, midline $11\ cm$

EXAMPLE 3 **A trapezium has area $84\ cm^2$, height $7\ cm$ and one parallel side $9\ cm$. Find the other.**

1. $$84 = \frac{9 + b}{2} \cdot 7$$

   Substitute what you know.

2. $$12 = \frac{9 + b}{2}$$

   Divide by 7.

3. $$24 = 9 + b$$

   Multiply by 2.

4. $$b = 15\ cm$$

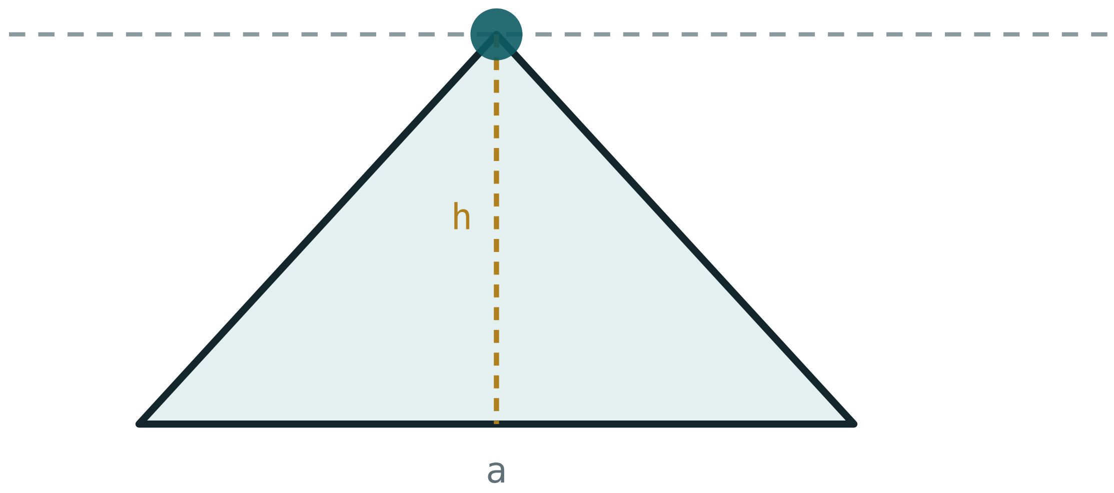
ANSWER

$15\ cm$

03 Interactive model

Change the base and height and check each formula against the readout.

Base, height and area



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rhombus	Romb	Ромб
Diagonal	Diagonal	Диагональ
Trapezium	Trapetsiya	Трапеция
Parallel sides	Parallel tomonlar	Основания трапеции
Midline	O‘rta chiziq	Средняя линия
Height of a trapezium	Trapetsiya balandligi	Высота трапеции
Kite	Deltoid	Дельтоид
Perpendicular diagonals	Perpendikulyar diagonallar	Перпендикулярные диагонали

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A rhombus with diagonals 10 and 8 has area:

- 80
- 40
- 18
- 20

A trapezium with parallel sides 6 and 10 and height 4 has area:

- 64
- 32
- 40
- 16

The midline of a trapezium with parallel sides 7 and 13 is:

- 20
- 10
- 6
- 91

A square of diagonal 6 has area:

- 36
- 18
- 12
- 9

A rhombus of area 60 has one diagonal 10. The other is:

- 6
- 12
- 120
- 30

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A rhombus has diagonals 6 and 8. Find its area.*

   ANSWER *24*

2. *A trapezium has parallel sides 5 and 9 and height 4. Find its area.*

   ANSWER *28*

3. *Find the midline of a trapezium with parallel sides 4 and 10.*

   ANSWER *7*

4. *A square has diagonal 10. Find its area.*

   ANSWER *50*

5. *A rhombus has diagonals 14 and 10. Find its area.*

   ANSWER *70*

6. *A trapezium has midline 8 and height 5. Find its area.*

   ANSWER *40*

7. *A rhombus has area 48 and one diagonal 12. Find the other.*

   ANSWER *8*

Medium · the standard question

7 PROBLEMS

1. *A rhombus has diagonals 16 and 30. Find its area and its side.*

   ANSWER *240*, side *17*

2. *A trapezium has area 96, height 8 and one parallel side 10. Find the other.*

   ANSWER *14*

3. *A rhombus of side 13 has one diagonal 24. Find its area.*

   ANSWER other diagonal *10*, area *120*

4. *A trapezium has parallel sides 12 and 18 and area 105. Find its height.*

   ANSWER *7*

5. *A kite has perpendicular diagonals 9 and 14. Find its area.*

   ANSWER *63*

6. *A rhombus has side 10 and height 8. Find its area.*

   ANSWER *80*

7. *A trapezium has midline 11 and area 77. Find its height.*

   ANSWER *7*

Hard · stretch and reason

7 PROBLEMS

1. *A rhombus has perimeter 40 and one diagonal 12. Find its area.*

   ANSWER side *10*, other diagonal *16*, area *96*

2. *An isosceles trapezium has parallel sides 10 and 22 and legs 10. Find its area.*

   ANSWER $h = 8$, area *128*

3. *A trapezium has parallel sides in the ratio 2 : 5, height 6 and area 63. Find both sides.*

   ANSWER *6* and *15*

4. *A rhombus of area 120 has side 13. Find both diagonals.*

   ANSWER *10* and *24*

5. *Show that the formula for a trapezium gives the parallelogram formula when $a = b$.*

   ANSWER $S = \frac{a + a}{2} \cdot h = a \cdot h$ — the trapezium becomes a parallelogram.

6. *A trapezium is divided by a diagonal into two triangles. Find the ratio of their areas if the parallel sides are 6 and 10.*

   ANSWER $3 : 5$ — the two triangles share the height

7. *The diagonals of a quadrilateral are perpendicular, of lengths 12 and 9. Must its area be 54?*

   ANSWER Yes — cutting along the diagonals gives four right triangles whose areas sum to $\frac{1}{2} d_1 d_2$, whatever the quadrilateral's shape, provided the diagonals meet inside it.

07 Homework

Homework — 5 problems

Geometry 8, Темы 49–50, pp. 114–118. Check every trapezium answer against $b \cdot h$.

1. *A rhombus has diagonals 18 and 24. Find its area and its side.*
2. *A trapezium has parallel sides 7 and 15 and height 6. Find its area.*
3. *A trapezium has area 120, height 10 and one parallel side 9. Find the other.*

4. *A square has diagonal 14. Find its area.*

5. *A rhombus has perimeter 52 and one diagonal 24. Find its area.*

Home / Grade 8 / Geometry / Quarter 3

The area of a polygon

Any polygon at all: cut it into triangles, add the pieces, done.

LESSON 50 | 1 × 40 MIN | GEOMETRY 8, TEMA 51 | STAGE 9 · 7.2

LESSON CLOCK

5' | 12' | 12' | 7' | 4'

Cut a pentagon into triangles	5 min
Decomposition	12 min
Compound shapes	12 min
Regular polygons	7 min
Homework	4 min

LEARNING OBJECTIVES

- Decompose a polygon into triangles and rectangles.
- Find the area of a compound shape by adding or subtracting.
- Find the area of a regular polygon from its apothem and perimeter.
- Find areas from coordinates on squared paper.

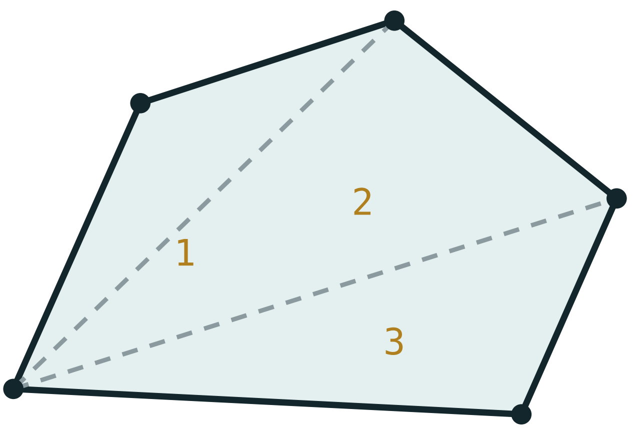
TEXTBOOK

National	Тема 51, pp. 119–121
Cambridge	Learner’s Book pp. 149–155
Workbook	Workbook 7.2

01 Explanation

Cut it into pieces you know

Any polygon can be cut into triangles by drawing diagonals from one vertex. By the additivity of area, the whole is the sum of the parts:



split into triangles, then add the areas

Diagonals from one vertex cut a polygon into triangles; add their areas to get the whole.

An n-gon splits into $n - 2$ triangles this way — the same count that gave the angle sum $180°(n - 2)$ in Quarter I. The two facts have the same source.

TWO STRATEGIES

Adding: cut the shape into rectangles and triangles and total them.
Subtracting: draw the smallest rectangle that contains the shape, then take away the corner pieces.
On squared paper the second is usually faster.

Regular polygons

A regular n-gon splits into n congruent triangles meeting at the centre. Each has base a (a side) and height r (the **apothem** — the distance from the centre to a side):

$$S = n \cdot \frac{1}{2} a r = \frac{1}{2} P r$$

where $P = na$ is the perimeter. A regular hexagon of side 6 has apothem $3\sqrt{3} \approx 5.2$, so $S = \frac{1}{2} \cdot 36 \cdot 5.2 \approx 93.5$.

APOTHEM, NOT RADIUS

The apothem reaches the **middle of a side**; the radius reaches a **vertex**. They are different lengths, and only the apothem belongs in this formula.

02 Worked examples

EXAMPLE 1

An L-shape is a 10×6 rectangle with a 4×3 rectangle removed from one corner. Find its area and perimeter.

1. $10 \cdot 6 = 60$

   The whole rectangle.

2. $4 \cdot 3 = 12$

   The piece removed.

3. $60 - 12 = 48$

   The area.

4. $P = 2(10 + 6) = 32$

   The perimeter is unchanged — the cut adds two edges and removes two of the same total length.

ANSWER

Area 48, perimeter 32

EXAMPLE 2 **Find the area of the polygon with vertices $(0;0)$, $(6;0)$, $(6;4)$, $(3;7)$, $(0;4)$.**

1. Split into a rectangle and a triangle.
   The horizontal line $y = 4$ does it.
2. *rectangle* $6 \times 4 = 24$
3. *triangle: base* 6, *height* 3
   From $y = 4$ up to $y = 7$.
4. $\frac{1}{2} \cdot 6 \cdot 3 = 9$
5. $24 + 9 = 33$

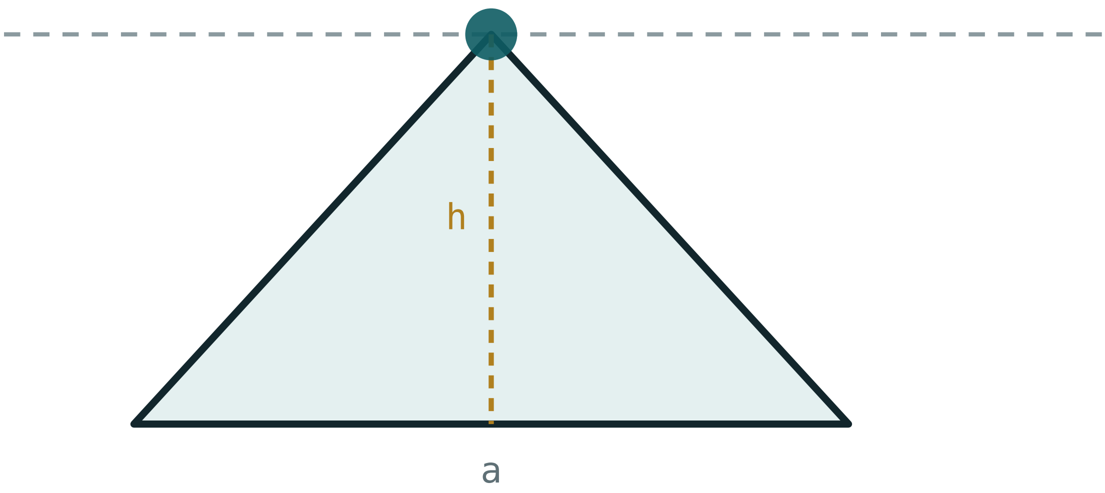
ANSWER

33

03 Interactive model

Change the base and height of one piece and watch how the total responds.

One piece of a decomposition



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Polygon	Ko‘pburchak	Многоугольник
Decomposition	Bo‘laklarga ajratish	Разбиение
Compound shape	Murakkab figura	Составная фигура
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Apothem	Apofema	Апофема
Enclosing rectangle	O‘rab turgan to‘rtburchak	Описанный прямоугольник
Subtraction method	Ayirish usuli	Метод вычитания
Cross-section	Kesim	Сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A pentagon splits into how many triangles from one vertex?

- 3
- 4
- 5
- 2

An L-shape is a 8×5 rectangle less a 3×2 corner. Its area is:

- 40
- 34
- 46
- 6

For a regular polygon, S equals:

- $P \cdot r$
- $\frac{1}{2} P r$
- $P + r$
- $n \cdot r$

The apothem of a regular polygon reaches:

- a vertex
- the middle of a side
- the opposite vertex
- the circumcircle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *An L-shape is a 6×4 rectangle less a 2×2 corner. Find its area.*

   ANSWER *20*

2. *A hexagon splits into how many triangles from one vertex?*

   ANSWER *4*

3. *Find the area of a shape made of a 5×3 rectangle and a triangle of base 5, height 2.*

   ANSWER *20*

4. *A regular polygon has perimeter 24 and apothem 4. Find its area.*

   ANSWER *48*

5. *An L-shape is a 9×5 rectangle less a 4×2 corner. Find its area.*

   ANSWER *37*

6. *Find the area of a triangle with vertices (0;0), (4;0), (0;3).*

   ANSWER *6*

7. *A regular hexagon has perimeter 36 and apothem 5.2. Find its area.*

   ANSWER ≈ 93.6

Medium · the standard question

7 PROBLEMS

1. *Find the area of the polygon (0;0), (8;0), (8;3), (4;6), (0;3).*

   ANSWER *36*

2. *A rectangle 12×8 has a 4×4 square cut from two opposite corners. Find the area left.*

   ANSWER *64*

3. *A T-shape is a 10×2 bar on top of a 4×6 column. Find its area.*

   ANSWER *44*

4. *A regular octagon has perimeter 40 and apothem 6. Find its area.*

   ANSWER *120*

5. *Find the area of the quadrilateral (0;0), (5;0), (7;4), (2;4).*

   ANSWER *20*

6. *A cross is made of five 3×3 squares. Find its area and perimeter.*

   ANSWER Area *45*, perimeter *36*

7. *A trapezium sits on a rectangle: rectangle 8×3, trapezium with parallel sides 8 and 4, height 3. Find the total area.*

   ANSWER $24 + 18 = 42$

Hard · stretch and reason

7 PROBLEMS

1. *Find the area of the pentagon (0;0), (6;0), (8;4), (4;8), (0;4).*

   ANSWER *44*

2. *A regular hexagon has side 8. Find its apothem and area.*

   ANSWER $4\sqrt{3} \approx 6.93$, area ≈ 166.3

3. *A room 6 m by 5 m has a 2 m by 1.5 m alcove added. Find the floor area and the cost of tiling at 120 000 so‘m/m².*

   ANSWER *33 m², 3 960 000 so‘m*

4. *A rectangle 10 × 6 has a triangle of base 10 and height 6 removed. Find the area left, and explain the answer.*

   ANSWER $60 - 30 = 30$ — the triangle is exactly half the rectangle

5. *Find the area of the quadrilateral (1;1), (7;2), (6;6), (2;5).*

   ANSWER *20*

6. *A regular polygon has 12 sides of length 5 and apothem 9.3. Find its area.*

   ANSWER ≈ 279

7. *Explain why cutting a polygon into triangles always gives the same total area, whichever cuts you use.*

   ANSWER Additivity says the parts of any one decomposition sum to the whole, and the whole does not depend on how it was cut — so every valid decomposition gives the same total.

07 Homework

Homework — 5 problems

Geometry 8, Тема 51, pp. 119–121. Draw the cutting lines before calculating.

1. *An L-shape is a 12 × 7 rectangle less a 5 × 3 corner. Find its area.*
2. *Find the area of the polygon (0;0), (7;0), (7;4), (3;7), (0;4).*
3. *A regular polygon has perimeter 30 and apothem 5. Find its area.*

4. *A T-shape is a 12×3 bar on a 4×5 column. Find its area.*

5. *A regular hexagon has side 10. Find its apothem and area.*

Practical · The circumference and area of a circle

Cambridge insert: the one shape with no straight sides, and the number π that measures it.

LESSON 51 | 1 × 40 MIN | GEOMETRY 8, ТЕМА 52 | STAGE 9 · 7.1

LESSON CLOCK

6' | 12' | 12' | 6' | 4'

Measure a round lid: C ÷ d	6 min
Circumference	12 min
Area	12 min
Arcs and sectors	6 min
Homework	4 min

LEARNING OBJECTIVES

- Use $C = 2\pi r$ and $S = \pi r^2$.
- Find the radius from a given circumference or area.
- Find the area of a sector and the length of an arc.
- Solve practical problems involving circles.

TEXTBOOK

National	Тема 52, pp. 122–125
Cambridge	Learner's Book pp. 142–148
Workbook	Workbook 7.1

01 Explanation

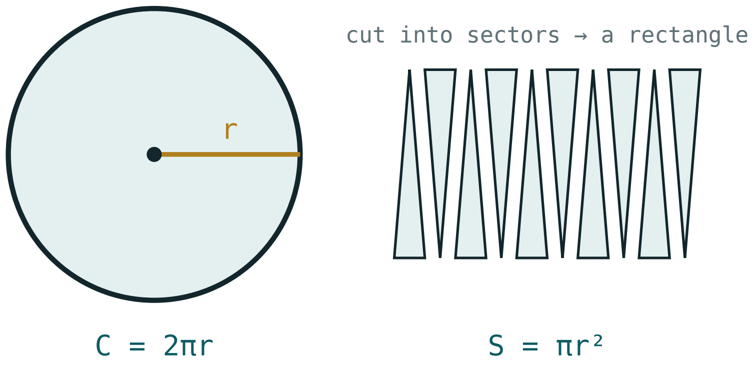
π, and the two formulas

Measure the circumference of any round object and divide by its diameter. Every time, for every circle, you get the same number:

$$\pi = \frac{C}{d} \approx 3.14159\ldots$$

CIRCUMFERENCE AND AREA

$$C = 2\pi r = \pi d \mid S = \pi r^2$$



Cut the circle into thin sectors and interleave them: the result approaches a rectangle of base πr and height r.

Why πr^2? Cut the circle into many thin sectors and lay them alternately up and down. The result is nearly a rectangle of height r and base half the circumference, πr — giving $\pi r \cdot r = \pi r^2$. The thinner the sectors, the better the fit.

RADIUS OR DIAMETER?

$S = \pi r^2$ needs the **radius**. Given a diameter of 10, the area is $\pi \cdot 5^2 = 25\pi$, not 100π — a factor of four out.

Arcs, sectors and rings

A sector of angle α is the fraction $\frac{\alpha}{360°}$ of the whole circle, so both its arc and its area are that same fraction:

$$\ell = \frac{\alpha}{360°} \cdot 2\pi r \;\mid\; S_{sector} = \frac{\alpha}{360°} \cdot \pi r^2$$

A quarter circle of radius 8: arc $\frac{1}{4} \cdot 16\pi = 4\pi \approx 12.6$, area $\frac{1}{4} \cdot 64\pi = 16\pi \approx 50.3$.

A **ring** (annulus) between radii R and r has area $\pi(R^2 - r^2)$ — the big disc less the small one, by additivity.

Give answers exactly in terms of π when possible, and only round at the very end.

02 Worked examples

EXAMPLE 1 **A circle has radius *7 cm*. Find its circumference and area, taking $\pi \approx 3.14$.**

1. $C = 2\pi \cdot 7 = 14\pi$
2. ≈ 43.96 cm
3. $S = \pi \cdot 49 = 49\pi$
4. ≈ 153.86 cm^2

ANSWER

$C \approx 44$ cm, $S \approx 153.9$ cm^2

EXAMPLE 2 **A circular pond has area *78.5 m²*. Find its radius and the length of fence needed round it.**

1. $78.5 = 3.14 \cdot r^2$
2. $r^2 = 25$, $r = 5$ m
3. $C = 2 \cdot 3.14 \cdot 5$
4. $= 31.4$ m

ANSWER

$r = 5$ m, fence 31.4 m

EXAMPLE 3

A circular running track has an inner radius of *30 m* and is *4 m* wide. Find the area of the track surface.

1. $R = 34,\ r = 30$

2. $S = \pi(34^2 - 30^2)$

   The ring formula.

3. $= \pi(1156 - 900) = 256\pi$

4. $\approx 804\ m^2$

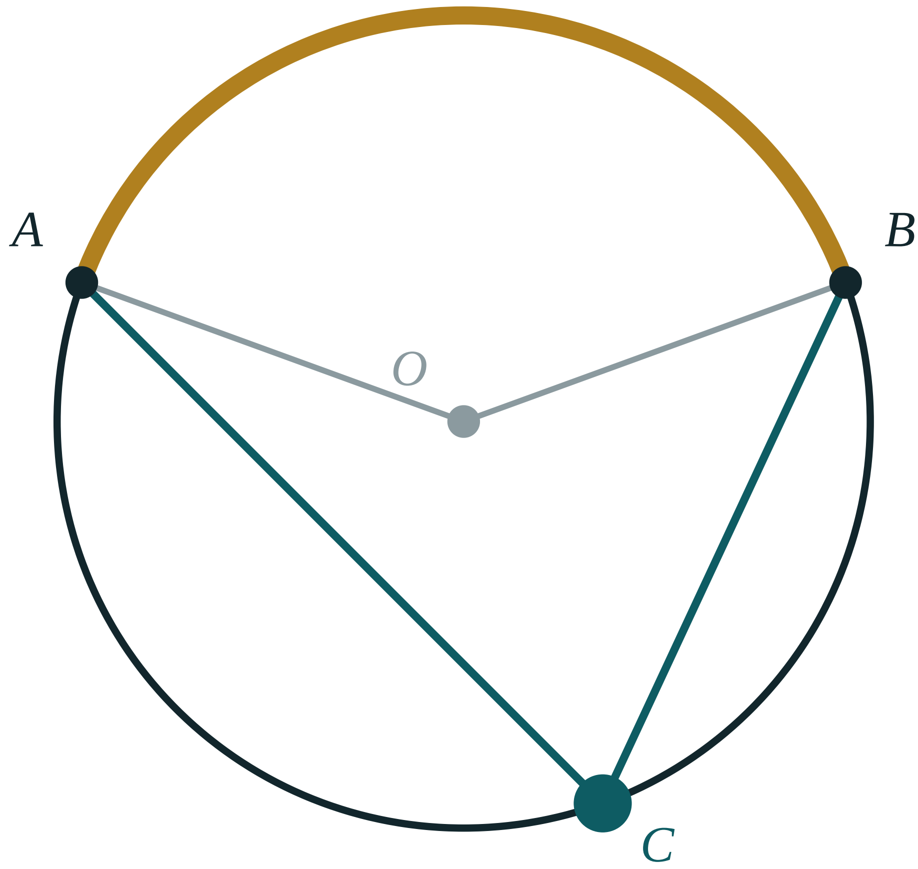
ANSWER

$256\pi \approx 804\ m^2$

03 Interactive model

Explore the circle by dragging the radius; compare the circumference and area readouts.

Around the circle



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Circle	Aylana	Окружность
Circumference	Aylana uzunligi	Длина окружности
Radius	Radius	Радиус
Diameter	Diametr	Диаметр
Pi (π)	Pi soni	Число π
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Semicircle	Yarim aylana	Полуокружность
Annulus (ring)	Halqa	Кольцо

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A circle of radius 5 has circumference:

- 5π
- 10π
- 25π
- 20π

A circle of radius 5 has area:

- 10π
- 25π
- 5π
- 50π

A circle of diameter 12 has area:

- 144π
- 36π
- 12π
- 24π

A quarter circle of radius 4 has area:

- 16π
- 4π
- 8π
- 2π

If the radius doubles, the area:

- doubles
- triples
- quadruples
- stays the same

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the circumference of a circle of radius 3 (in terms of π).*

   ANSWER 6π

2. *Find the area of a circle of radius 3 (in terms of π).*

   ANSWER 9π

3. *Find the area of a circle of radius 10.*

   ANSWER $100\pi \approx 314$

4. *Find the circumference of a circle of diameter 14.*

   ANSWER $14\pi \approx 44$

5. *Find the area of a semicircle of radius 6.*

   ANSWER 18π

6. *A circle has circumference 20π. Find its radius.*

   ANSWER 10

7. *Find the area of a quarter circle of radius 2.*

   ANSWER π

Medium · the standard question

7 PROBLEMS

1. *A circle has area 49π. Find its radius and circumference.*

   ANSWER $r = 7$, $C = 14\pi$

2. *Find the area of a sector of radius 6 and angle 60°.*

   ANSWER 6π

3. *Find the arc length of a sector of radius 9 and angle 40°.*

   ANSWER 2π

4. *A circular table of diameter 1.2 m is covered. Find the area of cloth needed.*

   ANSWER $0.36\pi \approx 1.13\ m^2$

5. *Find the area of the ring between radii 10 and 6.*

   ANSWER 64π

6. *A wheel of radius 35 cm makes one turn. How far does it travel?*

   ANSWER $70\pi \approx 220$ cm

7. *A circle has circumference 31.4 m. Find its area ($\pi \approx 3.14$).*

   ANSWER $r = 5$, area $78.5\ m^2$

Hard · stretch and reason

7 PROBLEMS

1. *A square of side 10 has a circle inscribed in it. Find the area between them.*

   ANSWER $100 - 25\pi \approx 21.5$

2. *A circle is inscribed in a square of side 10, and a square is inscribed in that circle. Find the smaller square's area.*

   ANSWER 50 — its diagonal is the circle's diameter, 10

3. *A running track is 400 m round with two straights of 100 m and two semicircular ends. Find the radius of the ends.*

   ANSWER $2\pi r = 200$, so $r \approx 31.8$ m

4. *A wheel of diameter 70 cm turns 500 times. How far does it travel, in metres?*

   ANSWER $500 \cdot 0.7\pi \approx 1100$ m

5. *A sector has area 12π and radius 6. Find its angle.*

   ANSWER $120°$

6. *A circular pizza of diameter 40 cm is cut into 8 equal slices. Find the area of one slice.*

   ANSWER $50\pi \approx 157$ cm^2

7. *Two circles have radii in the ratio 2 : 3. Find the ratio of their areas, and explain.*

   ANSWER $4 : 9$ — area depends on the square of the radius, so lengths in ratio k give areas in ratio k^2

07 Homework

Homework — 5 problems

Geometry 8, Тема 52, pp. 122–125. Leave answers in terms of π unless told otherwise.

1. *Find the circumference and area of a circle of radius 8.*
2. *A circle has area 144π. Find its radius and circumference.*
3. *Find the area of a sector of radius 10 and angle 72°.*

4. *Find the area of the ring between radii 12 and 9.*
5. *A bicycle wheel has diameter 66 cm. How far does it travel in 200 turns?*

Control work 3 · Coordinates, vectors and area

The Quarter III assessment, and the four errors it produces every year.

LESSON 52 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 53–54 | STAGE 9 · 7.1–7.2, 13.1–13.2

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess coordinates, distance, vectors and all the area formulas.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National	Темы 53–54, pp. 126–127
Cambridge	Learner's Book pp. 142–155, 284–296
Workbook	Workbook 7.1–7.2

01 Explanation

The paper (40 minutes)

Two variants of eight tasks. Tasks 1–5 are worth 1 mark, tasks 6–8 are worth 2:

- task 1 · the midpoint of a segment
- task 2 · the distance between two points
- task 3 · the equation of a circle from centre and radius
- task 4 · AB and $|AB|$ from two points
- task 5 · the area of a parallelogram or a triangle
- task 6 · the area of a trapezium or a rhombus
- task 7 · a compound shape
- task 8 · a coordinate proof

Marking note: the four errors below account for almost every lost mark in this paper.

The four errors

WORK ON MISTAKES

1. **Start minus end** in a vector — the answer comes out with both signs reversed.
2. **The slant side used as the height** of a parallelogram or trapezium.

3. **The trapezium formula without the halving** — an answer exactly twice too big.
4. **Radius and diameter swapped**, or r^2 read off as r in the circle equation.

Each pupil writes the error number beside every lost mark and re-solves that task in full.

02 Worked examples

EXAMPLE 1 **Find the error: for $A(1; 2)$ and $B(5; 7)$, $AB = (-4; -5)$**

1. The subtraction went the wrong way round.
   AB means "from A to B".

2. $AB = (5 - 1; 7 - 2)$
   End minus start.

3. $AB = (4; 5)$
   The answer given is BA.

ANSWER

$(4; 5)$

EXAMPLE 2 **Find the error: a trapezium with parallel sides *6* and *10* and height *4* has area *64***

1. The halving was left out.
   $(6 + 10) \cdot 4 = 64$ is the parallelogram, not the trapezium.

2. $$S = \frac{6 + 10}{2} \cdot 4$$

3. $= 8 \cdot 4 = 32$
   Check: less than $10 \cdot 4 = 40$ ✓

ANSWER

32

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: for $A(3; 1)$, $B(8; 4)$, $AB = (-5; -3)$

1. AB

runs **from A to B**

· The subtraction is end minus start.

2. $(8 - 3; 4 - 1)$

3. $= (5; 3)$

What was written is BA.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Midpoint	O‘rta nuqta	Середина
Distance	Masofa	Расстояние
Vector	Vektor	Вектор
Height	Balandlik	Высота
Trapezium	Trapetsiya	Трапеция
Units of area	Yuza birliklari	Единицы площади

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For $A(2; 5)$ and $B(7; 9)$, AB is:

- $(-5; -4)$
- $(5; 4)$
- $(9; 14)$
- $(4; 5)$

A trapezium with parallel sides 5 and 9 and height 6 has area:

- 84
- 42
- 27
- 70

The radius of $(x - 1)^2 + (y - 2)^2 = 36$ is:

- 36
- 6
- 18
- 12

A parallelogram with sides 7 and 4 and an angle of $30°$ has area:

- 28
- 14
- 11
- 7

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Find the midpoint of (1; 3) and (7; 9).*

   ANSWER *(4; 6)*

2. ***Task 2.*** *Find the distance between (0; 0) and (5; 12).*

   ANSWER *13*

3. ***Task 3.*** *Write the equation of the circle, centre (2; −3), radius 5.*

   ANSWER $(x - 2)^2 + (y + 3)^2 = 25$

4. ***Task 4.*** *Find AB and |AB| for A(1; 1), B(4; 5).*

   ANSWER *(3; 4), 5*

5. ***Task 5.*** *Find the area of a triangle with base 12 and height 5.*

   ANSWER *30*

6. ***Task 6.*** *Find the area of a trapezium with parallel sides 6 and 10 and height 5.*

   ANSWER *40*

7. ***Task 7.*** *An L-shape is a 9×6 rectangle less a 3×2 corner. Find its area.*

   ANSWER *48*

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Find the midpoint of* $(-4; 6)$ *and* $(2; -2)$.

   ANSWER $(-1; 2)$

2. ***Task 2.*** *Find the distance between* $(-1; 2)$ *and* $(3; 5)$.

   ANSWER 5

3. ***Task 3.*** *State the centre and radius of* $(x + 4)^2 + (y - 1)^2 = 49$.

   ANSWER $(-4; 1), r = 7$

4. ***Task 4.*** *Find* AB *and* $|AB|$ *for* $A(-2; 3)$, $B(4; 11)$.

   ANSWER $(6; 8), 10$

5. ***Task 5.*** *Find the area of a parallelogram with base 11 and height 6.*

   ANSWER 66

6. ***Task 6.*** *Find the area of a rhombus with diagonals 10 and 12.*

   ANSWER 60

7. ***Task 8.*** *Show that* $(0;0)$, $(5;0)$, $(5;4)$, $(0;4)$ *is a rectangle.*

   ANSWER Both diagonals have midpoint $(2.5; 2)$ and both equal $\sqrt{41}$.

Hard · stretch and reason

7 PROBLEMS

1. *Find the fourth vertex of a parallelogram with (1;2), (6;3), (8;8).*

   ANSWER $(3;\ 7)$

2. *A rhombus has perimeter 40 and one diagonal 16. Find its area.*

   ANSWER side 10, other diagonal 12, area 96

3. *A trapezium has area 90, height 9 and one parallel side 8. Find the other.*

   ANSWER 12

4. *Find the area of the polygon (0;0), (8;0), (8;5), (4;8), (0;5).*

   ANSWER 52

5. $a = (3;\ -2)$ *and* $b = (-1;\ 5)$*. Find* $2a - b$ *and its length.*

   ANSWER $(7;\ -9)$, $\sqrt{130}$

6. *A circle has centre (2; 3) and passes through (6; 6). Find its equation and its area.*

   ANSWER $(x-2)^2 + (y-3)^2 = 25$, area 25π

7. *Show that (1;1), (5;3), (7;7), (3;5) is a parallelogram, and find its area.*

   ANSWER Diagonals share the midpoint $(4;\ 4)$; area 12

07 Homework

Homework — 5 problems

Re-solve every task you lost marks on, with the error number written beside it.

1. Re-solve, in full, each task you lost a mark on.
2. Beside each, write the error number (1–4) that caused it.
3. *Find* AB *and* $|AB|$ *for* $A(3;\ -1)$, $B(9;\ 7)$.
4. *Find the area of a trapezium with parallel sides 9 and 15 and height 8.*
5. *State the centre and radius of* $(x-5)^2 + (y+2)^2 = 64$.